SAP projekt

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Uvod Projekt je napravljen u svrhu predmeta Statistička obrada podataka. Napravio ga je tim “Boksači”.

Ucitavanje potrebnih paketa

```
library(dplyr)
library(readr)
library(knitr)
library(ggplot2)
library(kableExtra)
```

```
diabetes <- read_csv("diabetes_dataset00.csv")
```

Prikažimo prvo osnovno o podacima.

```
cat("Dimenzije dataseta su:", dim(diabetes))
```

```
## Dimenzije dataseta su: 70000 34
```

```
cat("\nBroj redaka je:", nrow(diabetes))
```

```
##
## Broj redaka je: 70000
```

```
cat("\nBroj stupaca je:" , ncol(diabetes))
```

```
##
## Broj stupaca je: 34
```

IME STUPCA	Opis značenja
Target	Vrsta dijabetesa u pacijenta
Genetic Marker	Specifična sekvenca DNA kojom “se može” predvidjeti dijabetes
Autoantibodies	Prisutnost antitijela koja tijelo proizvede protiv vlastitih stanica
Family History	Je li pacijent ima povijest dijabetesa u obitelji
Environmental Factors	Okolišni faktori koji mogu utjecati na razvoj dijabetesa, npr. zagađenje zraka
Insulin Levels	Razina inzulina u tijelu, što je ključno za procjenu funkcije gušterače i upravljanje dijabetesom
Age	Dob osobe
BMI	Indeks tjelesne mase (BMI)
Physical Activity	Razina tjelesne aktivnosti
Dietary Habits	Prehrambene navike
Blood Pressure	Krvni tlak
Cholesterol Levels	Razina kolesterola u krvi
Waist Circumference	Obujam struka
Blood Glucose Level	Razina glukoze u krvi
Ethnicity	Etnicitet osobe

IME STUPCA	Opis značenja
Socioeconomic Factors	Socioekonomski faktori
Smoking Status	Status pušenja
Alcohol Consumption	Konzumacija alkohola
Glucose Tolerance Test	Test tolerancije na glukozu, koji se koristi za dijagnosticiranje dijabetesa ili predijabetesa
History of PCOS	Je li žena imala dijagnozu ili simptome policističnog sindroma jajnika (PCOS)
Previous Gestational Diabetes	Povijest gestacijskog dijabetesa
Pregnancy History	Povijest trudnoće
Weight Gain During Pregnancy	Povećanje tjelesne mase tijekom trudnoće
Pancreatic Health	Zdravlje gušterače
Pulmonary Function	Funkcija pluća
Cystic Fibrosis Diagnosis	Dijagnoza cistične fibroze
Steroid Use History	Povijest korištenja steroida
Genetic Testing	Postoje li genetske predispozicije za dijabetes
Neurological Assessments	Neurološke procjene
Liver Function Tests	Test funkcije jetre
Digestive Enzyme Levels	Razina probavnih enzima
Urine Test	Test urina
Birth Weight	Tjelesna masa pri rođenju
Early Onset Symptoms	Simptomi koji se javljaju u ranom stadiju dijabetesa

Deksriptivna analiza tablice diabetes: Kratki prikaz tablice

```
head(diabetes)
```

```
## # A tibble: 6 × 34
##   Target                      `Genetic Markers` Autoantibodies `Family History`
##   <chr>                       <chr>             <chr>          <chr>
## 1 Steroid-Induced Diabetes     Positive          Negative       No
## 2 Neonatal Diabetes Mellitus … Positive          Negative       No
## 3 Prediabetic                  Positive          Positive       Yes
## 4 Type 1 Diabetes              Negative          Positive       No
## 5 Wolfram Syndrome             Negative          Negative       Yes
## 6 LADA                         Positive          Negative       Yes
## # ℹ 30 more variables: `Environmental Factors` <chr>, `Insulin Levels` <dbl>,
## #   Age <dbl>, BMI <dbl>, `Physical Activity` <chr>, `Dietary Habits` <chr>,
## #   `Blood Pressure` <dbl>, `Cholesterol Levels` <dbl>,
## #   `Waist Circumference` <dbl>, `Blood Glucose Levels` <dbl>, Ethnicity <chr>,
## #   `Socioeconomic Factors` <chr>, `Smoking Status` <chr>,
## #   `Alcohol Consumption` <chr>, `Glucose Tolerance Test` <chr>,
## #   `History of PCOS` <chr>, `Previous Gestational Diabetes` <chr>, …
```

```
glimpse(diabetes)
```

```
## Rows: 70,000
## Columns: 34
## $ Target                          <chr> "Steroid-Induced Diabetes", "Neonatal …
## $ `Genetic Markers`               <chr> "Positive", "Positive", "Positive", "N…
## $ Autoantibodies                  <chr> "Negative", "Negative", "Positive", "P…
## $ `Family History`                <chr> "No", "No", "Yes", "No", "Yes", "Yes",…
## $ `Environmental Factors`         <chr> "Present", "Present", "Present", "Pres…
## $ `Insulin Levels`                <dbl> 40, 13, 27, 8, 17, 17, 29, 10, 47, 21,…
## $ Age                             <dbl> 44, 1, 36, 7, 10, 41, 30, 3, 47, 72, 6…
## $ BMI                             <dbl> 38, 17, 24, 16, 17, 26, 31, 18, 25, 24…
## $ `Physical Activity`             <chr> "High", "High", "High", "Low", "High",…
## $ `Dietary Habits`                <chr> "Healthy", "Healthy", "Unhealthy", "Un…
## $ `Blood Pressure`                <dbl> 124, 73, 121, 100, 103, 127, 115, 80, …
## $ `Cholesterol Levels`            <dbl> 201, 121, 185, 151, 146, 208, 237, 157…
## $ `Waist Circumference`           <dbl> 50, 24, 36, 29, 33, 32, 43, 29, 40, 36…
## $ `Blood Glucose Levels`          <dbl> 168, 178, 105, 121, 289, 142, 186, 206…
## $ Ethnicity                       <chr> "Low Risk", "Low Risk", "Low Risk", "L…
## $ `Socioeconomic Factors`         <chr> "Medium", "High", "Medium", "High", "L…
## $ `Smoking Status`                <chr> "Smoker", "Non-Smoker", "Smoker", "Smo…
## $ `Alcohol Consumption`           <chr> "High", "Moderate", "High", "Moderate"…
## $ `Glucose Tolerance Test`        <chr> "Normal", "Normal", "Abnormal", "Abnor…
## $ `History of PCOS`               <chr> "No", "Yes", "Yes", "No", "No", "No", …
## $ `Previous Gestational Diabetes` <chr> "No", "No", "No", "Yes", "Yes", "No", …
## $ `Pregnancy History`             <chr> "Normal", "Normal", "Normal", "Normal"…
## $ `Weight Gain During Pregnancy`  <dbl> 18, 8, 15, 12, 2, 11, 15, 4, 30, 33, 3…
## $ `Pancreatic Health`             <dbl> 36, 26, 56, 49, 10, 40, 62, 13, 91, 86…
## $ `Pulmonary Function`            <dbl> 76, 60, 80, 89, 41, 85, 64, 44, 71, 69…
## $ `Cystic Fibrosis Diagnosis`     <chr> "No", "Yes", "Yes", "Yes", "No", "Yes"…
## $ `Steroid Use History`           <chr> "No", "No", "No", "No", "No", "No", "Y…
## $ `Genetic Testing`               <chr> "Positive", "Negative", "Negative", "P…
## $ `Neurological Assessments`      <dbl> 3, 1, 1, 2, 1, 2, 3, 1, 3, 2, 3, 1, 1,…
## $ `Liver Function Tests`          <chr> "Normal", "Normal", "Abnormal", "Abnor…
## $ `Digestive Enzyme Levels`       <dbl> 56, 28, 55, 60, 24, 52, 96, 29, 74, 42…
## $ `Urine Test`                    <chr> "Ketones Present", "Glucose Present", …
## $ `Birth Weight`                  <dbl> 2629, 1881, 3622, 3542, 1770, 3835, 44…
## $ `Early Onset Symptoms`          <chr> "No", "Yes", "Yes", "No", "No", "Yes",…
```

Pošto imamo i kategorijske podatke na koje summary() funkcija neće imati klasičan odgovor napraviti će mo faktorizaciju kategorijskih stupaca

```
categorical_vars <- c(
  "Target", "Genetic Markers", "Autoantibodies",
  "Family History", "Environmental Factors",
  "Physical Activity", "Dietary Habits", "Ethnicity",
  "Socioeconomic Factors", "Smoking Status",
  "Alcohol Consumption", "Glucose Tolerance Test",
  "History of PCOS", "Previous Gestational Diabetes",
  "Pregnancy History", "Cystic Fibrosis Diagnosis",
  "Steroid Use History", "Genetic Testing",
  "Liver Function Tests", "Urine Test",
  "Early Onset Symptoms"
)

# Faktorizacija
diabetes[categorical_vars] <- lapply(diabetes[categorical_vars], as.factor)

s <- summary(diabetes)

library(pander)
pander(s, style = "rmarkdown")
```

Table continues below

Target	Genetic Markers
MODY : 5553	Negative:34899

Target	Genetic Markers
Secondary Diabetes : 5479	Positive:35101
Cystic Fibrosis-Related Diabetes (CFRD): 5464	NA
Type 1 Diabetes : 5446	NA
Neonatal Diabetes Mellitus (NDM) : 5408	NA
Wolcott-Rallison Syndrome : 5400	NA
(Other) :37250	NA

Table continues below

Autoantibodies	Family History	Environmental Factors	Insulin Levels
Negative:35058	No :34832	Absent :35088	Min. : 5.00
Positive:34942	Yes:35168	Present:34912	1st Qu.:13.00
NA	NA	NA	Median :19.00
NA	NA	NA	Mean :21.61
NA	NA	NA	3rd Qu.:28.00
NA	NA	NA	Max. :49.00
NA	NA	NA	NA

Table continues below

Age	BMI	Physical Activity	Dietary Habits
Min. : 0.00	Min. :12.00	High :23225	Healthy :35020
1st Qu.:14.00	1st Qu.:20.00	Low :23348	Unhealthy:34980
Median :31.00	Median :25.00	Moderate:23427	NA
Mean :32.02	Mean :24.78	NA	NA
3rd Qu.:49.00	3rd Qu.:29.00	NA	NA
Max. :79.00	Max. :39.00	NA	NA
NA	NA	NA	NA

Table continues below

Blood Pressure	Cholesterol Levels	Waist Circumference
Min. : 60.0	Min. :100.0	Min. :20.00
1st Qu.: 99.0	1st Qu.:163.0	1st Qu.:30.00
Median :113.0	Median :191.0	Median :34.00
Mean :111.3	Mean :194.9	Mean :35.05
3rd Qu.:125.0	3rd Qu.:225.0	3rd Qu.:39.00
Max. :149.0	Max. :299.0	Max. :54.00
NA	NA	NA

Table continues below

Blood Glucose Levels	Ethnicity	Socioeconomic Factors
Min. : 80.0	High Risk:34982	High :23304

Blood Glucose Levels	Ethnicity	Socioeconomic Factors
1st Qu.:121.0	Low Risk :35018	Low :23283
Median :152.0	NA	Medium:23413
Mean :160.7	NA	NA
3rd Qu.:194.0	NA	NA
Max. :299.0	NA	NA
NA	NA	NA

Table continues below

Smoking Status	Alcohol Consumption	Glucose Tolerance Test
Non-Smoker:34955	High :23246	Abnormal:35278
Smoker :35045	Low :23411	Normal :34722
NA	Moderate:23343	NA
NA	NA	NA
NA	NA	NA
NA	NA	NA
NA	NA	NA

Table continues below

History of PCOS	Previous Gestational Diabetes	Pregnancy History
No :35101	No :34965	Complications:34730
Yes:34899	Yes:35035	Normal :35270
NA	NA	NA
NA	NA	NA
NA	NA	NA
NA	NA	NA
NA	NA	NA

Table continues below

Weight Gain During Pregnancy	Pancreatic Health	Pulmonary Function
Min. : 0.0	Min. :10.00	Min. :30.00
1st Qu.: 7.0	1st Qu.:32.00	1st Qu.:63.00
Median :16.0	Median :46.00	Median :72.00
Mean :15.5	Mean :47.56	Mean :70.26
3rd Qu.:22.0	3rd Qu.:64.00	3rd Qu.:79.00
Max. :39.0	Max. :99.00	Max. :89.00
NA	NA	NA

Table continues below

Cystic Fibrosis Diagnosis	Steroid Use History	Genetic Testing
No :35135	No :35142	Negative:34685

Cystic Fibrosis Diagnosis	Steroid Use History	Genetic Testing
Yes:34865	Yes:34858	Positive:35315
NA	NA	NA
NA	NA	NA
NA	NA	NA
NA	NA	NA
NA	NA	NA

Table continues below

Neurological Assessments	Liver Function Tests	Digestive Enzyme Levels
Min. :1.000	Abnormal:34981	Min. :10.00
1st Qu.:1.000	Normal :35019	1st Qu.:31.00
Median :2.000	NA	Median :48.00
Mean :1.804	NA	Mean :46.42
3rd Qu.:2.000	NA	3rd Qu.:61.00
Max. :3.000	NA	Max. :99.00
NA	NA	NA

Urine Test	Birth Weight	Early Onset Symptoms
Glucose Present:17422	Min. :1500	No :35059
Ketones Present:17422	1st Qu.:2629	Yes:34941
Normal :17528	Median :3103	NA
Protein Present:17628	Mean :3097	NA
NA	3rd Qu.:3656	NA
NA	Max. :4499	NA
NA	NA	NA

1. Projektno pitanje

“Može li se na temelju zadanih parametara procijeniti dob pacijenta?”

Promatramo utjecaj samo jedne nezavisne varijable X na neku zavisnu varijablu Y. Graficki cemo to prikazati s scatter plotom. Za sada odabiremo par X varijabli za koje smatramo da bismo iz njih nesto mogli zakljuciti o starosti pacijenata tj. biramo za koje smatramo da imaju logicku vezu s dobi.

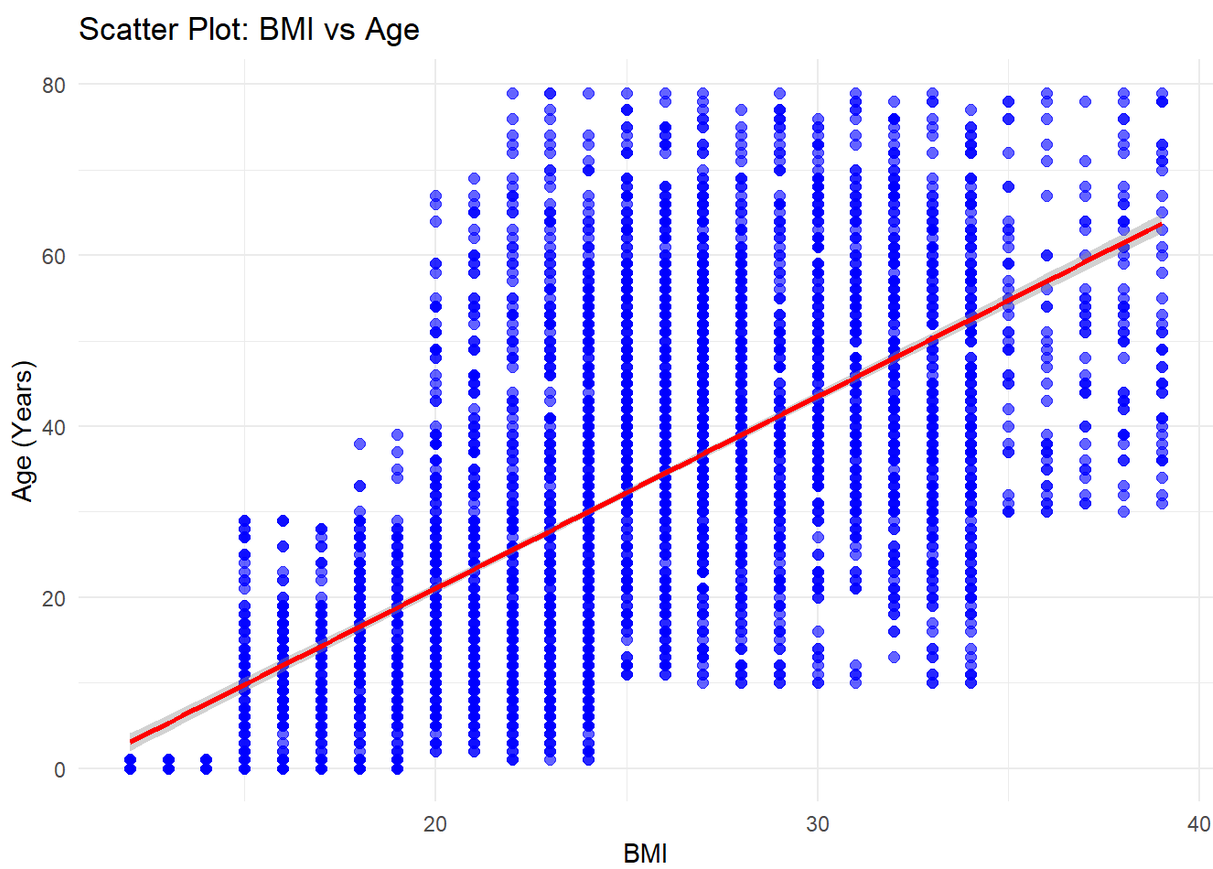
Neke varijable (i razlozi) koje cemo crtati scatter plotom su

- BMI (Indeks tjelesne mase): S dobi se često mijenja tjelesna masa i sastav tijela
- Blood Pressure (Krvni tlak): S godinama krvni tlak obično raste zbog starenja krvnih žila
- Physical Activity: Prirodno je da stariji ljudi se manje bave fizickom aktivnosti
- Cholesterol Levels (Razina kolesterola): Razine kolesterola mogu rasti s godinama
- Blood Glucose Level: Kako ljudi stare, osjetljivost na inzulin često opada, pa bi razine glukoze mogle biti više kod starijih pacijenata.
- Waist Circumference (Obujam struka):

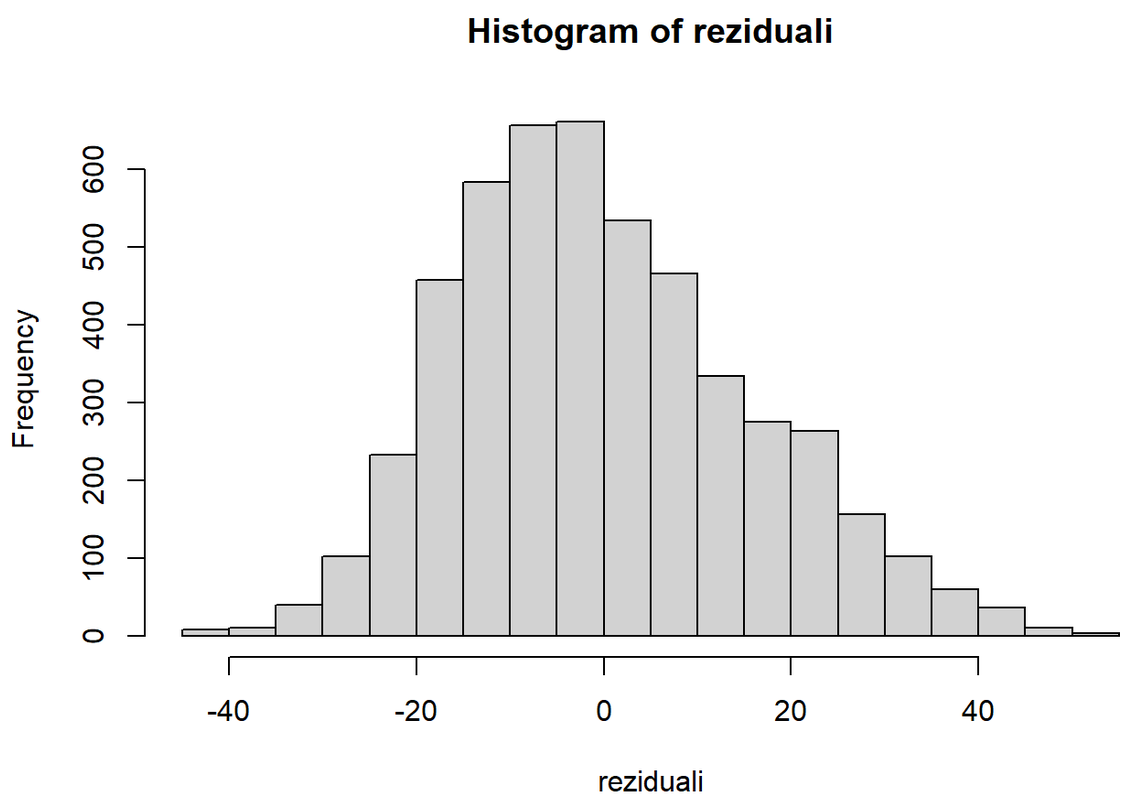
Scatter plot: BMI vs Age

```
# BMI
diab_red <- sample_n(diabetes, 5000) # reducirani skup, jer nam je previse tocaka


ggplot(diab_red, aes(x = BMI, y = Age)) +
  geom_point(color = "blue", alpha = 0.6, size = 2) +
  geom_smooth(method = "lm", color = "red", se = TRUE) + # Dodaje regresijsku crtu
  labs(title = "Scatter Plot: BMI vs Age",
       x = "BMI",
       y = "Age (Years)") +
  theme_minimal()
```

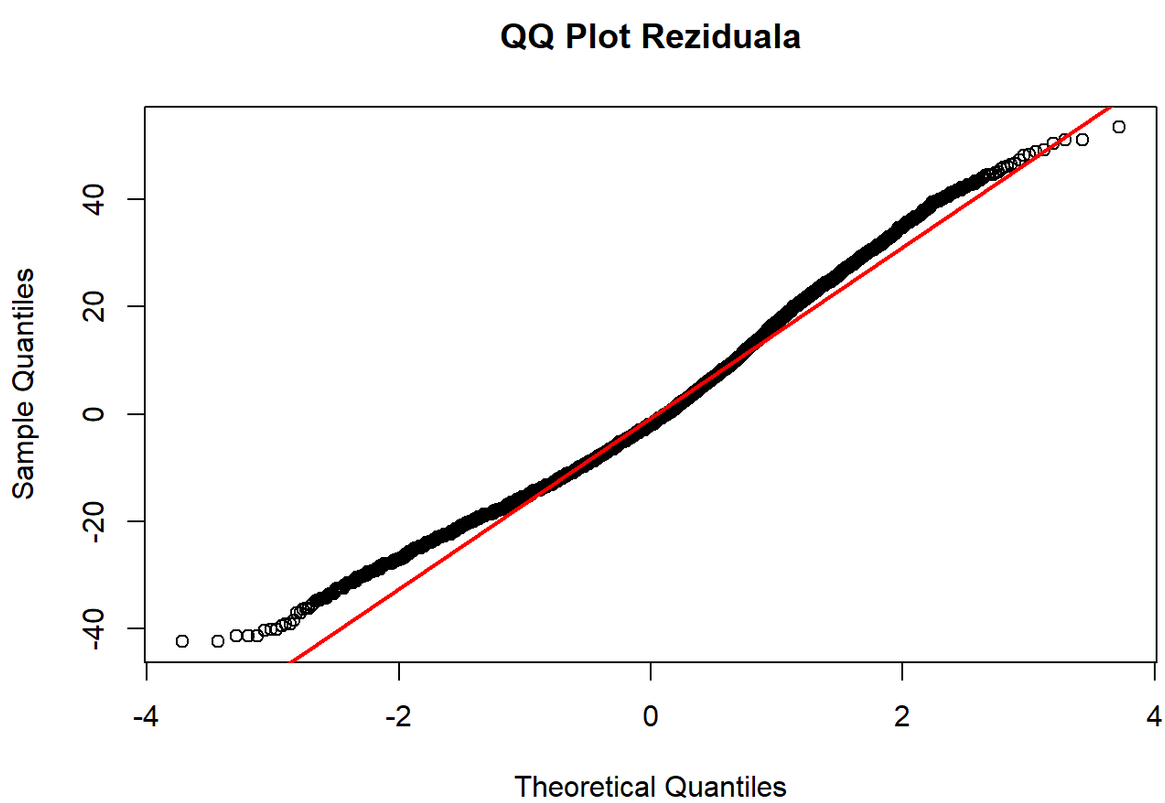
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
fit.BMI <-lm(Age ~ BMI, data = diab_red)


#provjeravamo normalnost reziduala - bitna pretpostavka
reziduali <- residuals(fit.BMI)



# uocavamo da nije previse narusena normalnost reziduala
hist(reziduali)
```



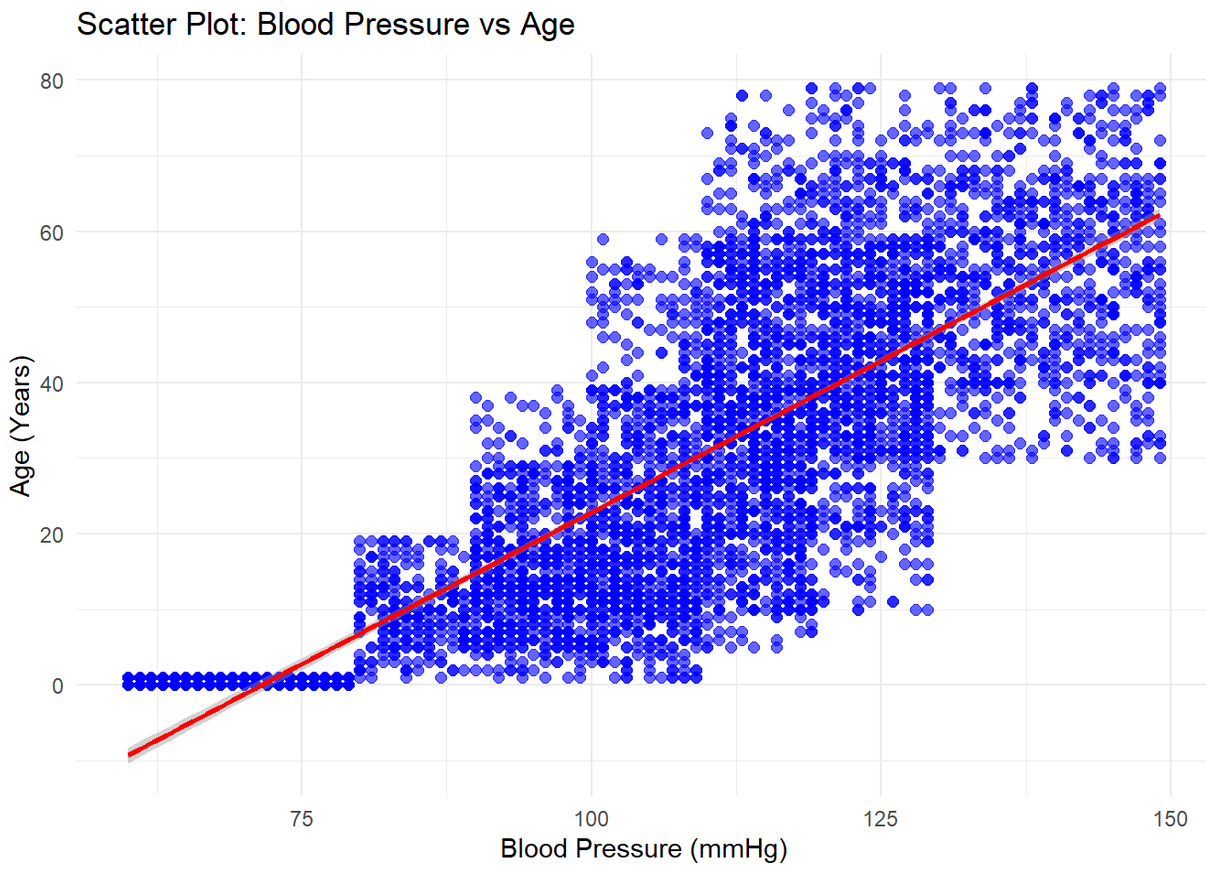

```
qqnorm(reziduali, main = "QQ Plot Reziduala")
qqline(reziduali, col = "red", lwd = 2)
```



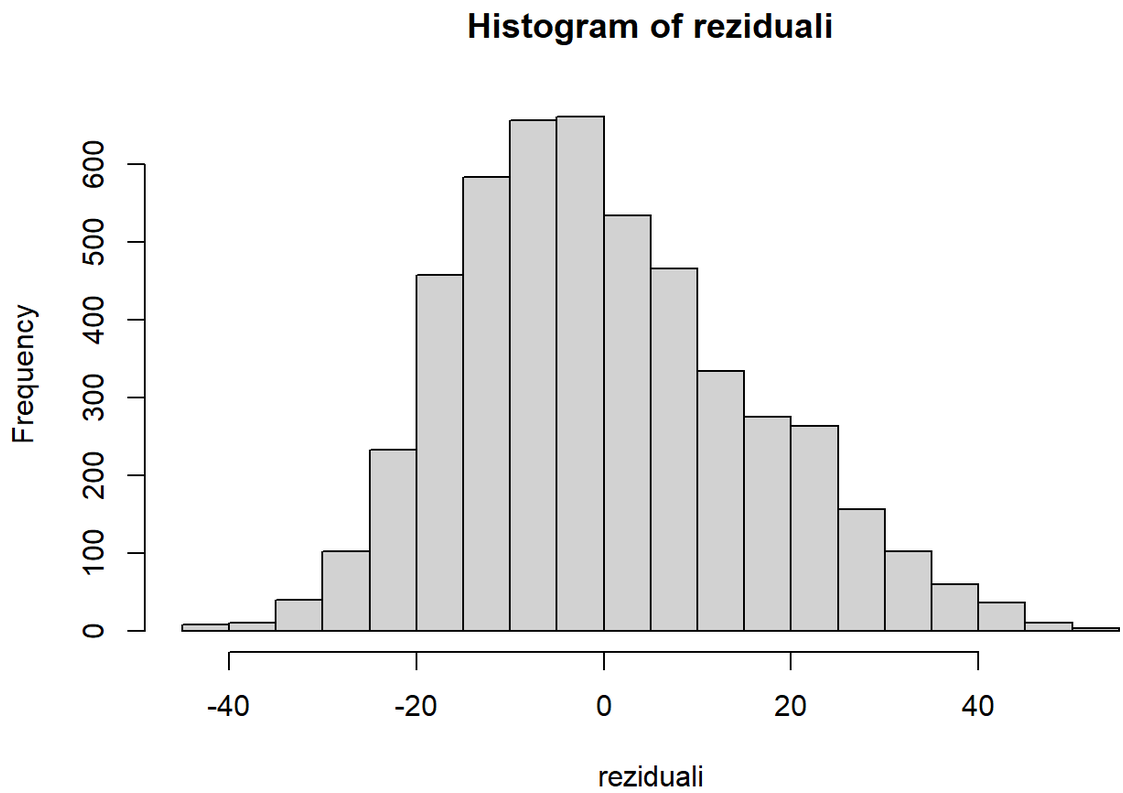
Scatter plot: Blood Pressure vs Age

```
ggplot(diab_red, aes(x = `Blood Pressure`, y = Age)) +
  geom_point(color = "blue", alpha = 0.6, size = 2) +
  labs(title = "Scatter Plot: Blood Pressure vs Age",
       x = "Blood Pressure (mmHg)",
       y = "Age (Years)") +
  geom_smooth(method = "lm", color = "red", se = TRUE) + # Dodaje regresijsku crtu
  # lm znaci linearni model
  theme_minimal()
```

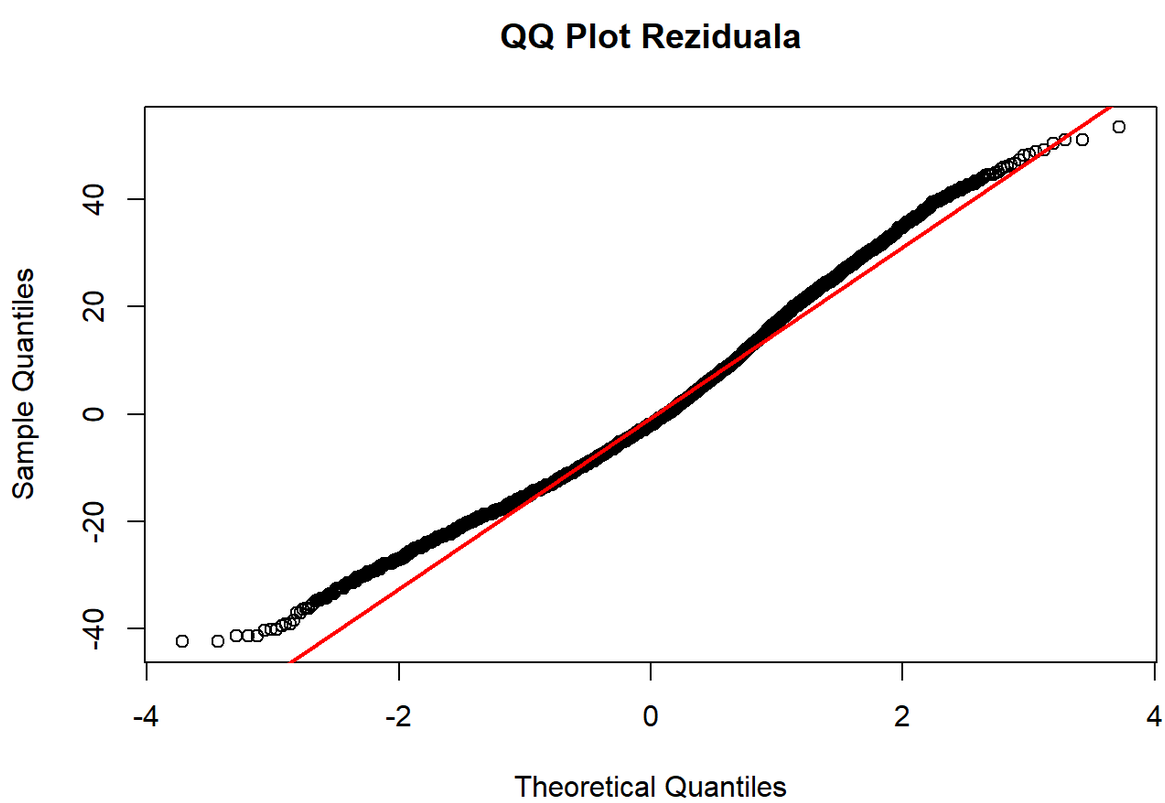
```
## `geom_smooth()` using formula = 'y ~ x'
```



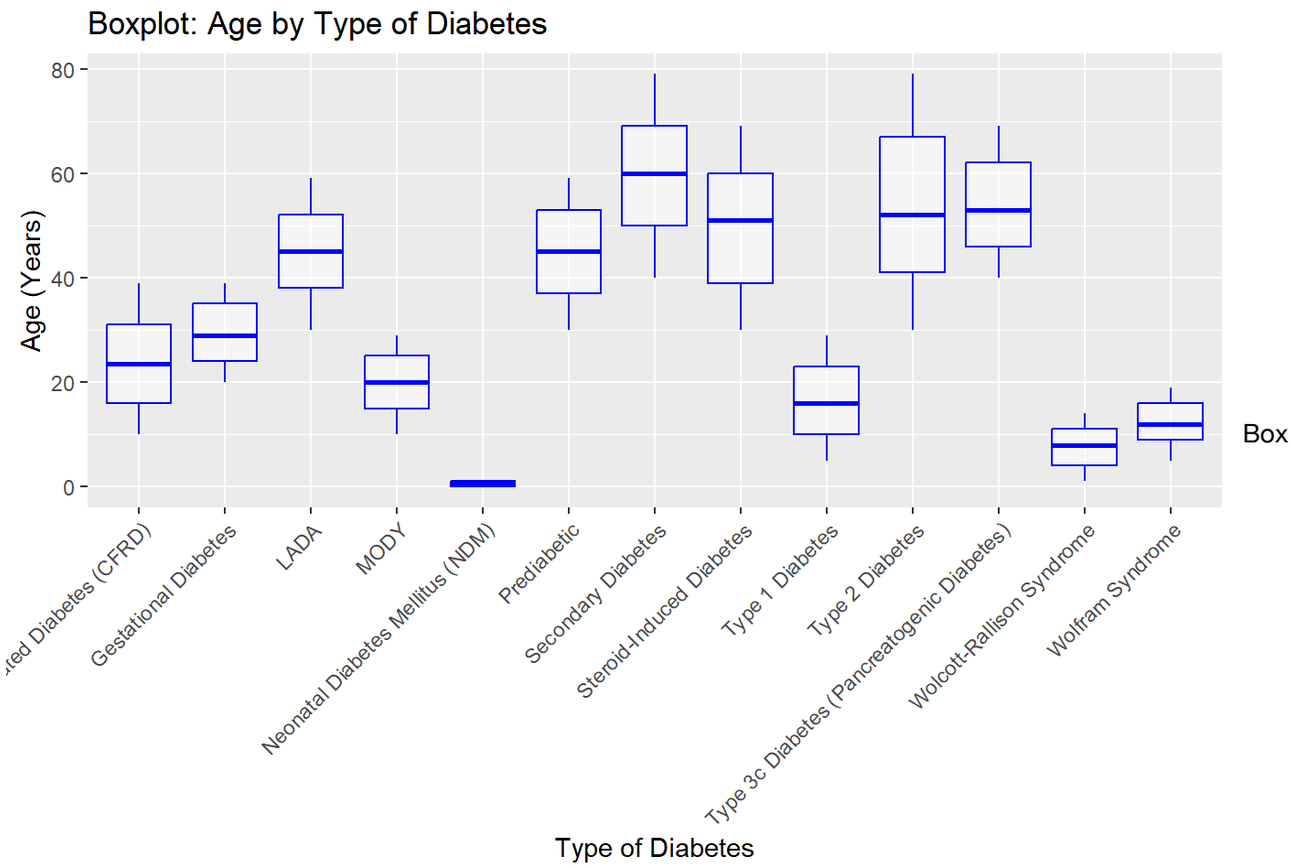

```
# uocavamo lagani trend da se povecavanjem godina povecava i krvni tlak


fit.bp <-lm(Age ~ BMI, data = diab_red)

#provjeravamo normalnost reziduala - bitna pretpostavka
reziduali <- residuals(fit.bp)



# uocavamo da nije previse narusena normalnost reziduala
hist(reziduali)
```




```
qqnorm(reziduali, main = "QQ Plot Reziduala")
qqline(reziduali, col = "red", lwd = 2)
```



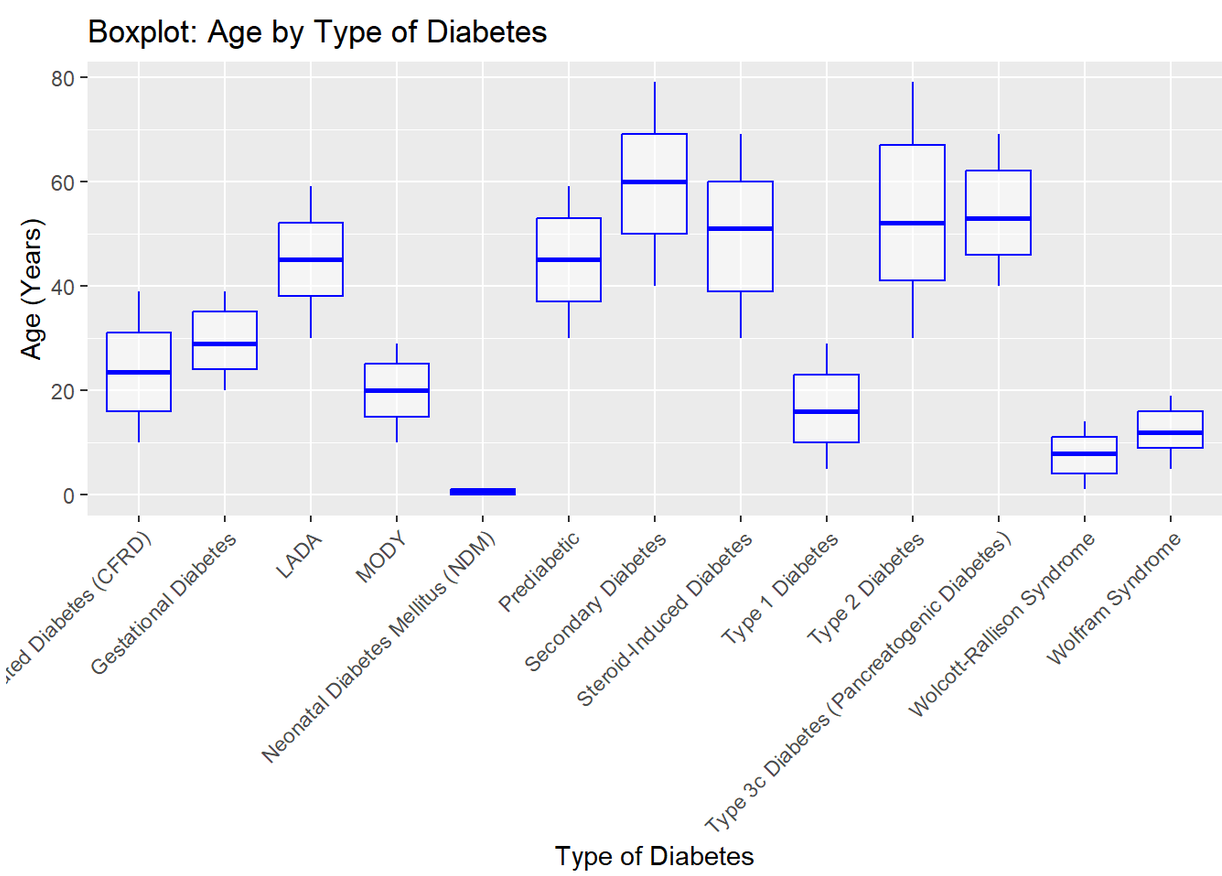

```
ggplot(diab_red, aes(x = `Target`, y = Age)) +
  geom_boxplot(color = "blue", alpha = 0.6) +
  labs(title = "Boxplot: Age by Type of Diabetes",
       x = "Type of Diabetes",
       y = "Age (Years)") +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```



Box

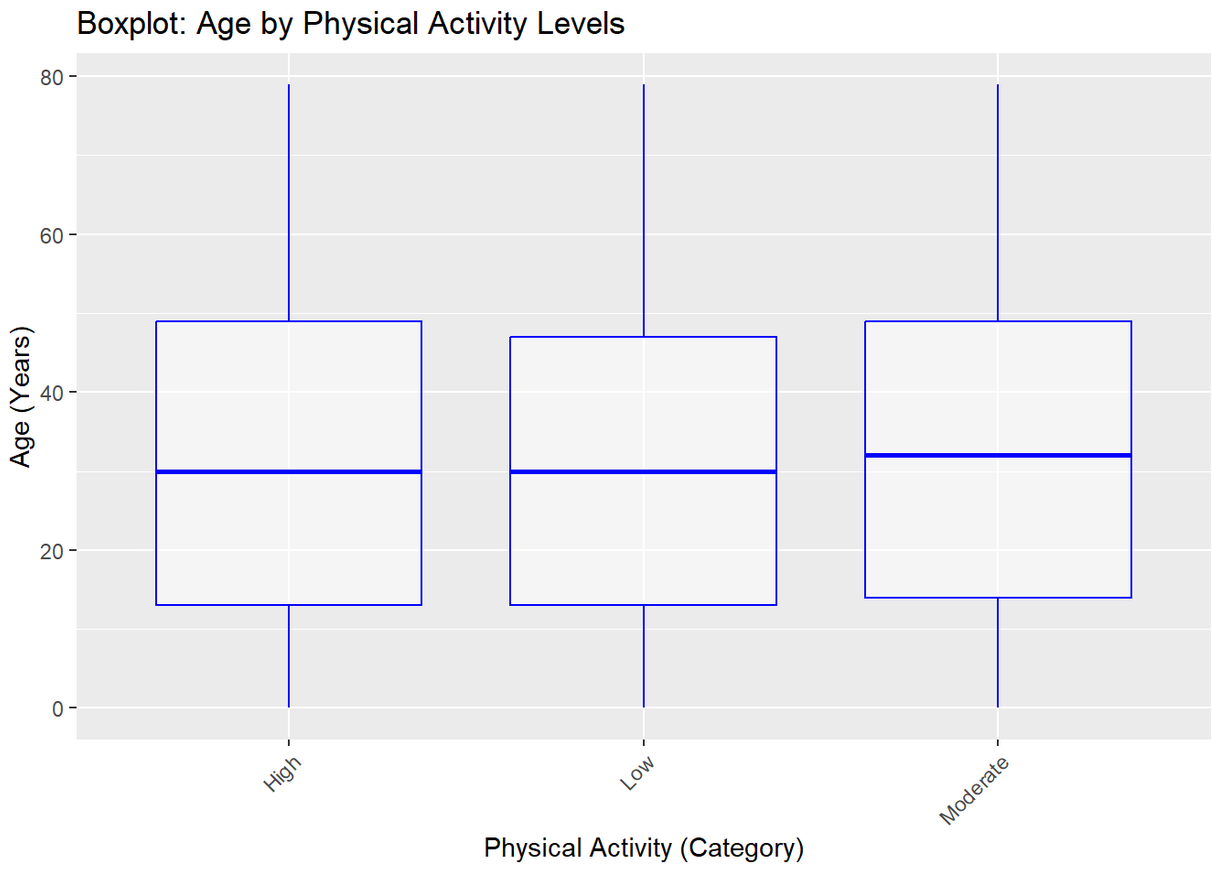
plot: Physical Activity vs Age

```
ggplot(diab_red, aes(x = `Target`, y = Age)) +
  geom_boxplot(color = "blue", alpha = 0.6) +
  labs(title = "Boxplot: Age by Type of Diabetes",
       x = "Type of Diabetes",
       y = "Age (Years)") +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```



Box plot: Physical Activity vs Age

```
ggplot(diab_red, aes(x = `Physical Activity`, y = Age)) +
  geom_boxplot(color = "blue", alpha = 0.6) +
  labs(title = "Boxplot: Age by Physical Activity Levels",
       x = "Physical Activity (Category)",
       y = "Age (Years)") +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```

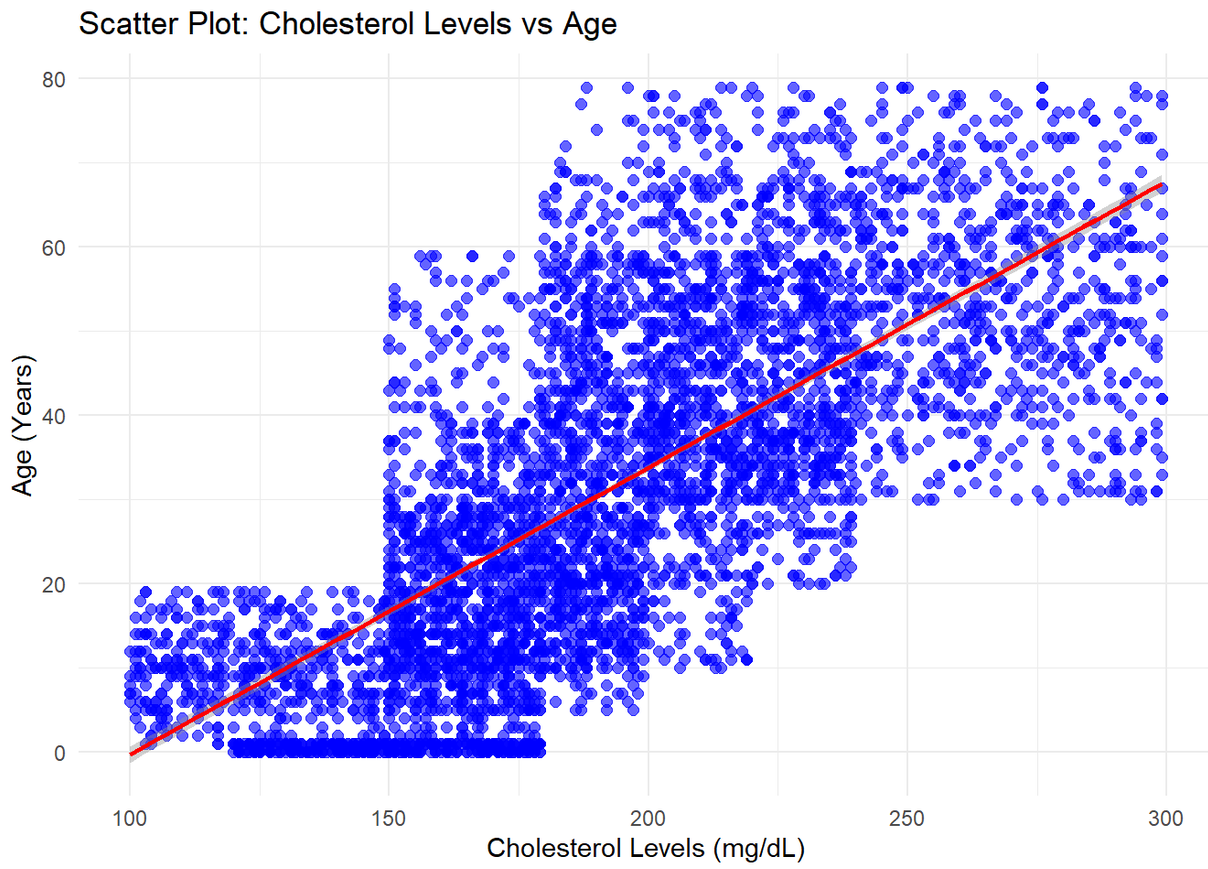


Ovaj

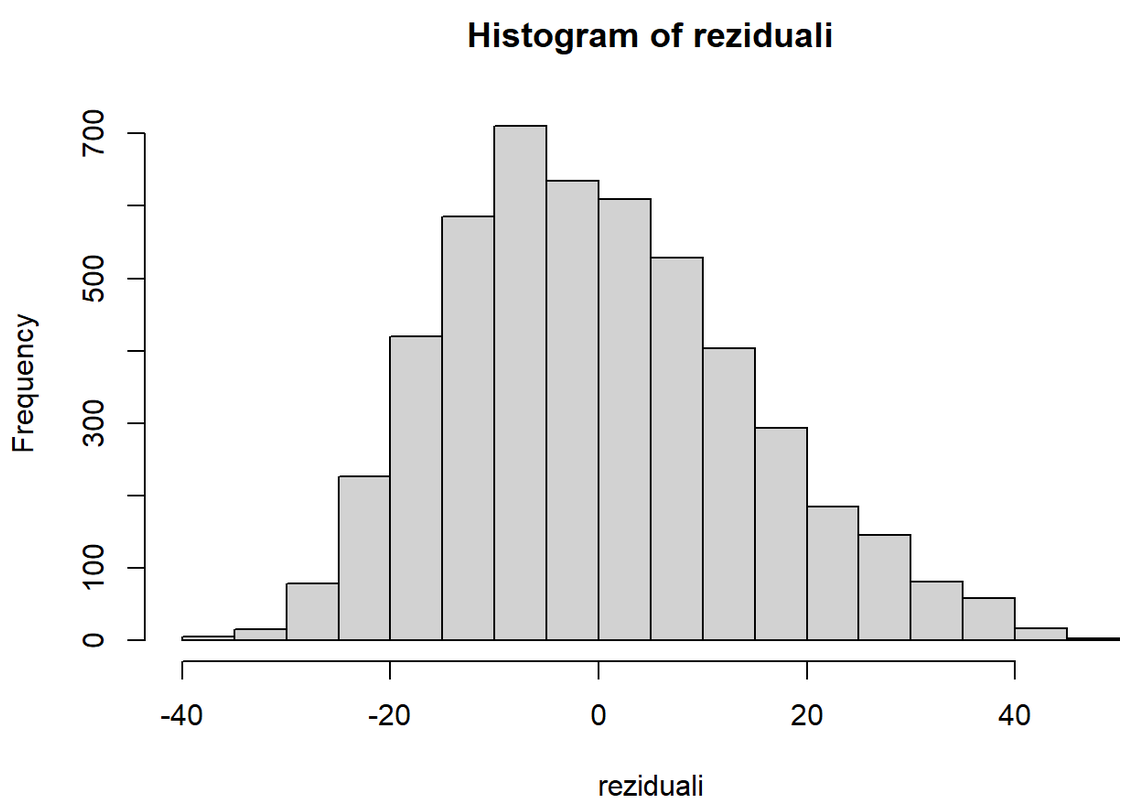
box plot nam jako malo govori o povezanosti između fizičke aktivnosti i godina

```
ggplot(diab_red, aes(x = `Cholesterol Levels`, y = Age)) +
  geom_point(color = "blue", alpha = 0.6, size = 2) +
  geom_smooth(method = "lm", color = "red", se = TRUE) + # Dodaje regresijsku crtu
  labs(title = "Scatter Plot: Cholesterol Levels vs Age",
       x = "Cholesterol Levels (mg/dL)",
       y = "Age (Years)") +
  theme_minimal()
```

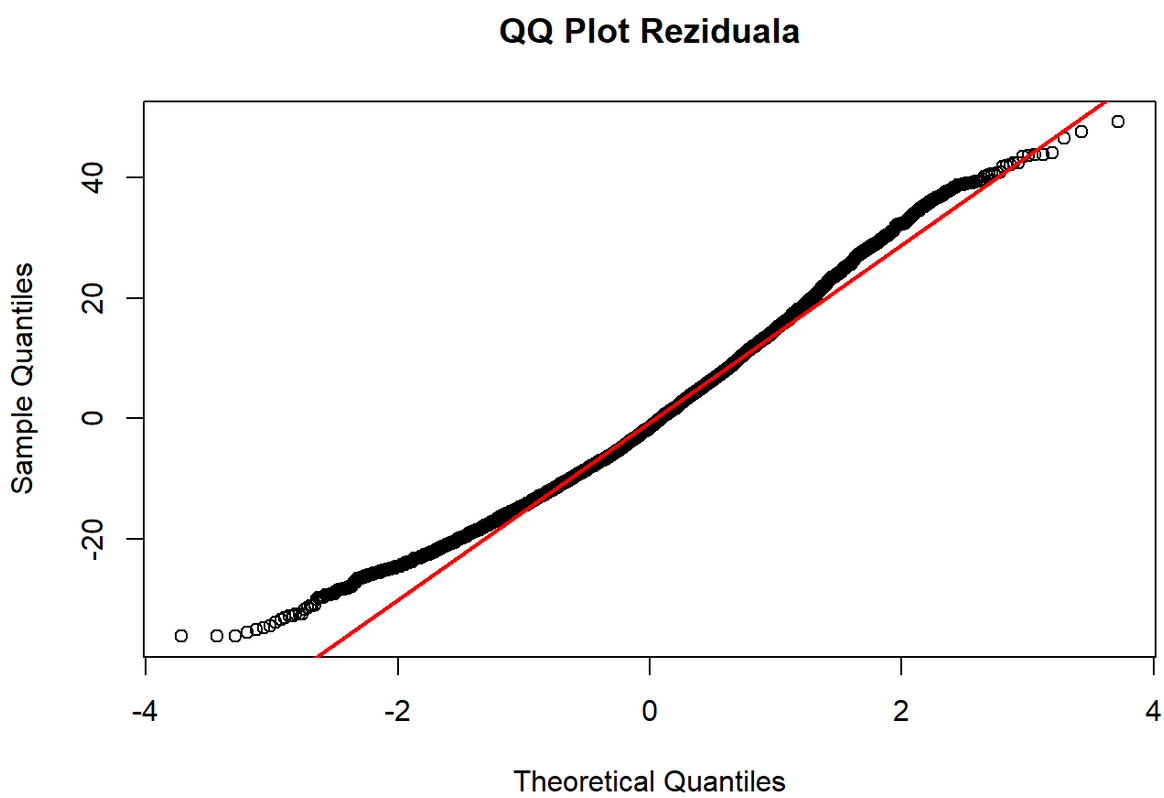
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
fit.ch <-lm(Age ~ `Cholesterol Levels`, data = diab_red)

#provjeravamo normalnost reziduala - bitna pretpostavka
reziduali <- residuals(fit.ch)



# uocavamo da nije previse narusena normalnost reziduala
hist(reziduali)
```

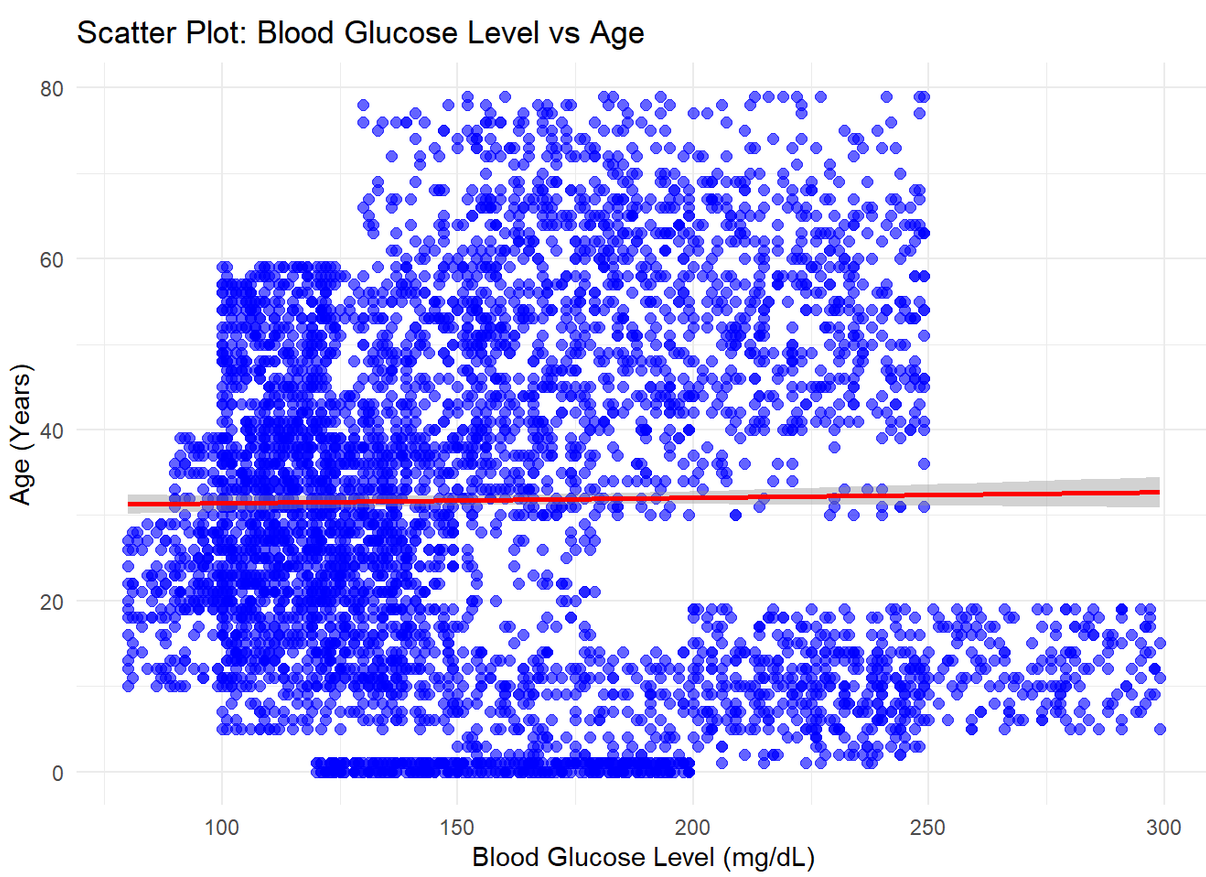



```
qqnorm(reziduali, main = "QQ Plot Reziduala")
qqline(reziduali, col = "red", lwd = 2)
```




```
ggplot(diab_red, aes(x = `Blood Glucose Levels`, y = Age)) +
  geom_point(color = "blue", alpha = 0.6, size = 2) +
  labs(title = "Scatter Plot: Blood Glucose Level vs Age",
       x = "Blood Glucose Level (mg/dL)",
       y = "Age (Years)") +
  geom_smooth(method = "lm", color = "red", se = TRUE) + # Dodaje regresijsku crtu
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

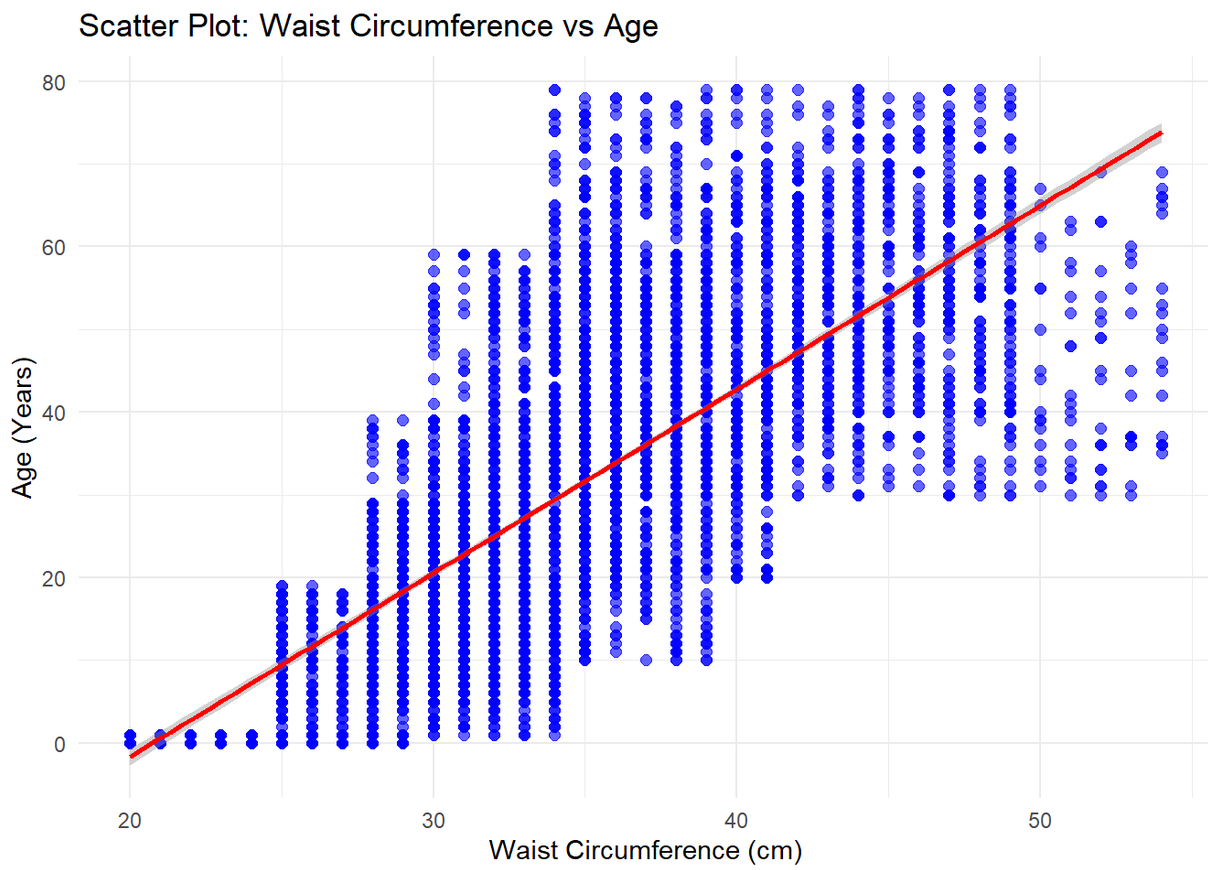


Ne uočavamo nikakvu linearnu poveznost - jako loš regresor

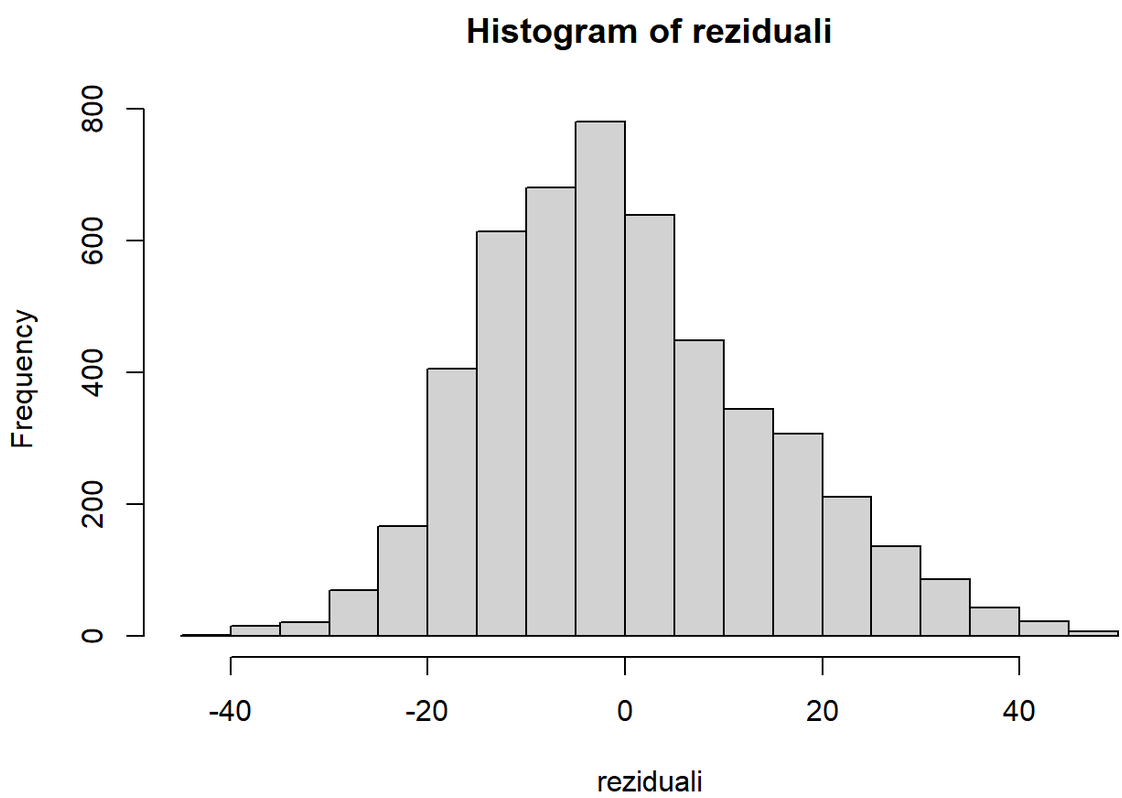
Scatter plot: Waist Circumference vs Age

```
ggplot(diab_red, aes(x = `Waist Circumference`, y = Age)) +
  geom_point(color = "blue", alpha = 0.6, size = 2) +
  labs(title = "Scatter Plot: Waist Circumference vs Age",
       x = "Waist Circumference (cm)",
       y = "Age (Years)") +
  geom_smooth(method = "lm", color = "red", se = TRUE) + # Dodaje regresijsku crtu
  theme_minimal()
```

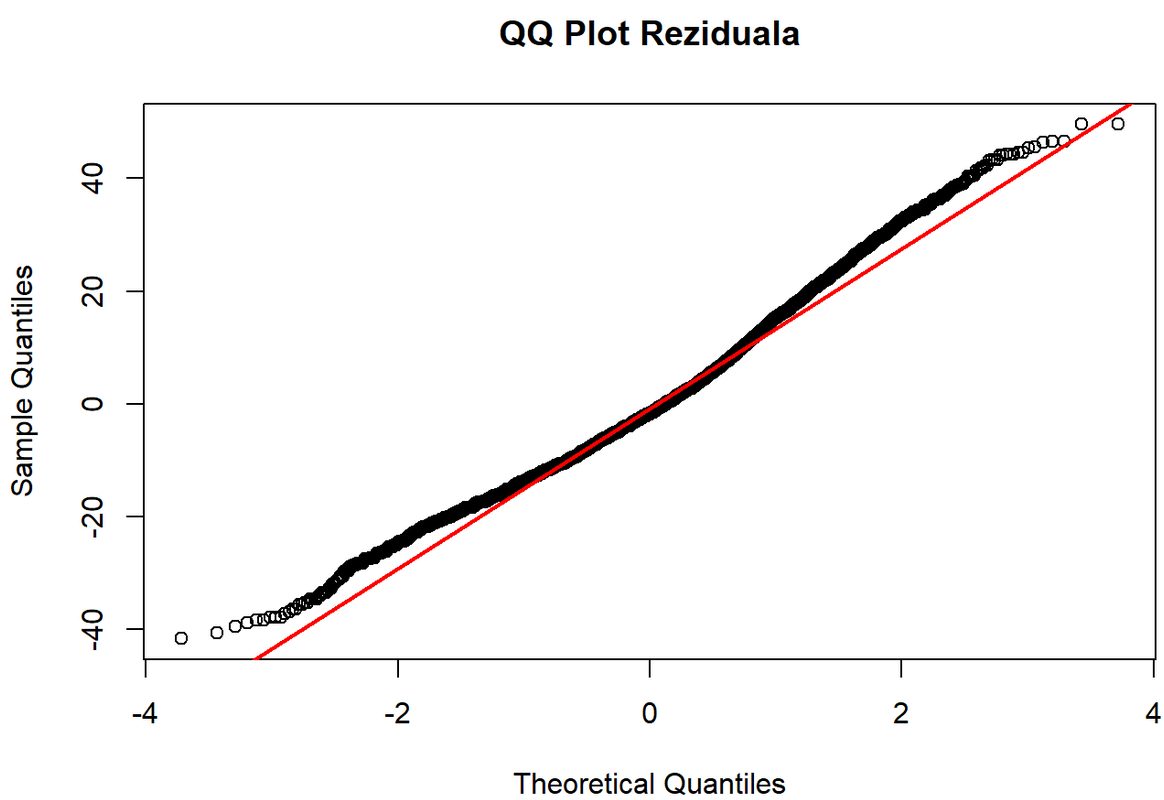
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
fit.ws <-lm(Age ~ `Waist Circumference`, data = diab_red)

#provjeravamo normalnost reziduala - bitna pretpostavka
reziduali <- residuals(fit.ws)


hist(reziduali)
```




```
qqnorm(reziduali, main = "QQ Plot Reziduala")
qqline(reziduali, col = "red", lwd = 2)
```



Uočavamo da nije previše narušena normalnost reziduala.

Dalje ćemo koristit Target,BMI, Blood Presure, Cholesterol i Waist kao regresore jer su njihovi grafovi u pocetnoj analizi dali najbolju linearnu vezu s varijablom dobi.

Sada ćemo viditi jesu li neki regresori kolerirani (koliko je jaka veza između parova regresora)

Korelacijski koeficijenti

Regresori: `Target BMI Blood Pressure Cholesterol Levels Waist Circumference`

```
korelacija <- cor(cbind(diab_red$Target,
                        diab_red$BMI,
                        diab_red$`Blood Pressure`,
                        diab_red$`Cholesterol Levels`,
                        diab_red$`Waist Circumference`)) # Samo potpuni podaci

# Dodavanje imena stupcima i redovima
colnames(korelacija) <- rownames(korelacija) <- c("Target",
                                                  "BMI",
                                                  "Blood Pressure",
                                                  "Cholesterol Levels",
                                                  "Waist Circumference")
korelacija
```

```
##                          Target        BMI Blood Pressure Cholesterol Levels
## Target               1.00000000 -0.1185633     0.03868907        -0.06304927
## BMI                 -0.11856330  1.0000000     0.65184262         0.59464883
## Blood Pressure       0.03868907  0.6518426     1.00000000         0.67779501
## Cholesterol Levels  -0.06304927  0.5946488     0.67779501         1.00000000
## Waist Circumference  0.03856185  0.6079267     0.70802792         0.66918774
##                     Waist Circumference
## Target                       0.03856185
## BMI                          0.60792672
## Blood Pressure               0.70802792
## Cholesterol Levels           0.66918774
## Waist Circumference          1.00000000
```

Iz ove tablice uocavamo zakljucke

BMI i Blood Pressure: Korelacija od 0.642 ukazuje na srednje jaku pozitivnu povezanost.

BMI i Cholesterol Levels: Korelacija od 0.595 je blizu granice, takođe ukazuje na povezanost.

BMI i Waist Circumference: Korelacija od 0.619 je također jaka.

Sada provodimo visestruku linearnu regresiju

Korištenje kategorijskih varijabli s više od dvije kategorije kao int vrijednosti u regresiji se ne preporuča za nominalne varijable. Dakle mi necemo koristiti Target varijablu, a njen boxplot je vec prikazan iznad i u primjeru 1.4. je razlozeno kako vrsta dijabetesa moze pokazivati jako dobro dob pacijenta.

```
# Višestruka regresija sa Age kao zavisnom varijablom
fit.multi <- lm(Age ~ BMI + `Blood Pressure` + `Cholesterol Levels` + `Waist Circumferen
ce`, data = diab_red)

# Prikaz rezultata regresije
summary(fit.multi)
```

```
##
## Call:
## lm(formula = Age ~ BMI + `Blood Pressure` + `Cholesterol Levels` +
##     `Waist Circumference`, data = diab_red)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -40.896  -7.598  -0.499   6.712  36.420
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -67.890823   0.967906  -70.14   <2e-16 ***
## BMI                     0.469267   0.037875   12.39   <2e-16 ***
## `Blood Pressure`        0.363726   0.013274   27.40   <2e-16 ***
## `Cholesterol Levels`    0.117384   0.005536   21.20   <2e-16 ***
## `Waist Circumference`   0.708610   0.037185   19.06   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.45 on 4995 degrees of freedom
## Multiple R-squared:  0.6954, Adjusted R-squared:  0.6951
## F-statistic:  2850 on 4 and 4995 DF,  p-value: < 2.2e-16
```

Uočavamo jako velike reziduale (40 godina na nekim procjenama je previše za godine(dob)). **No ipak medijan reziduala je blizu nule sto je poželjno.**

Svi koeficijenti su statistički značajni (p-vrijednosti < 0.001), što znači da su svi prediktori relevantni za objašnjenje starosti.

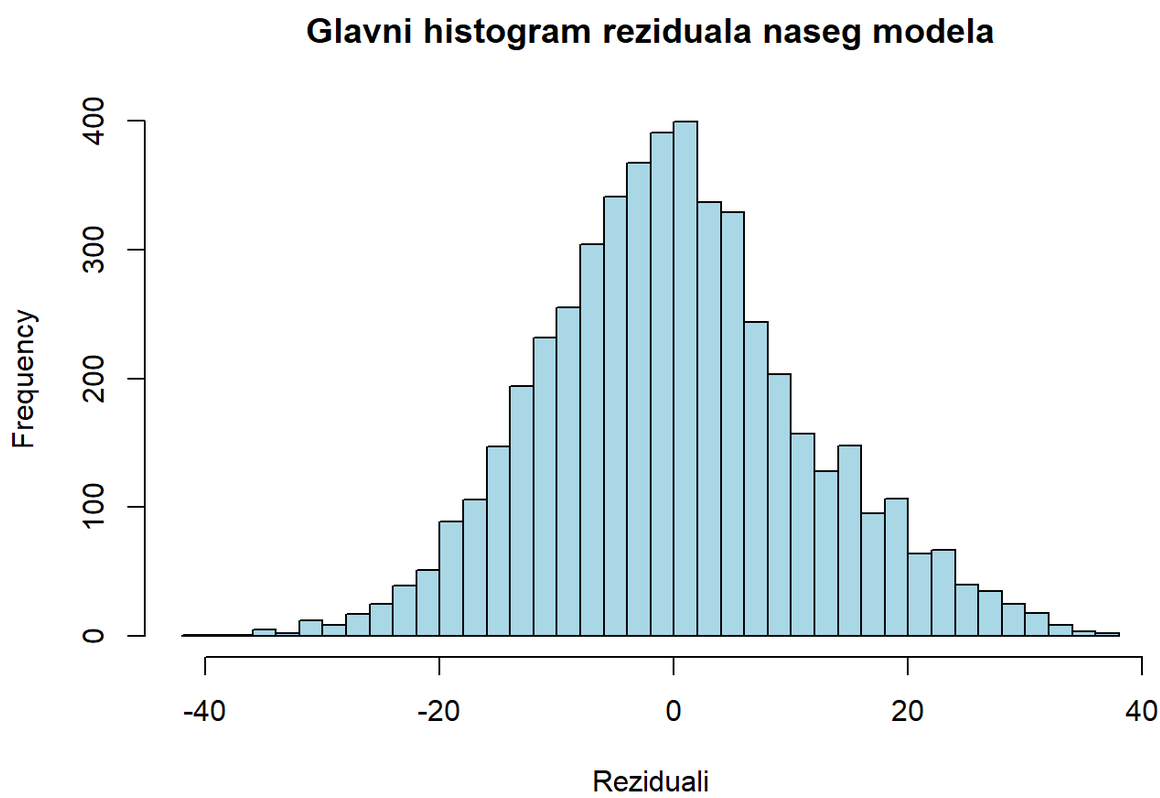
Nama je nulta hipoteza da je beta jednaka nuli, te nam ove p-vrijednosti govore da tu hipotezu možemo odbaciti!

Model objašnjava 70.35% varijance zavisne varijable Age, što ukazuje na dobru sposobnost modela da objasni starost na osnovu navedenih prediktora.

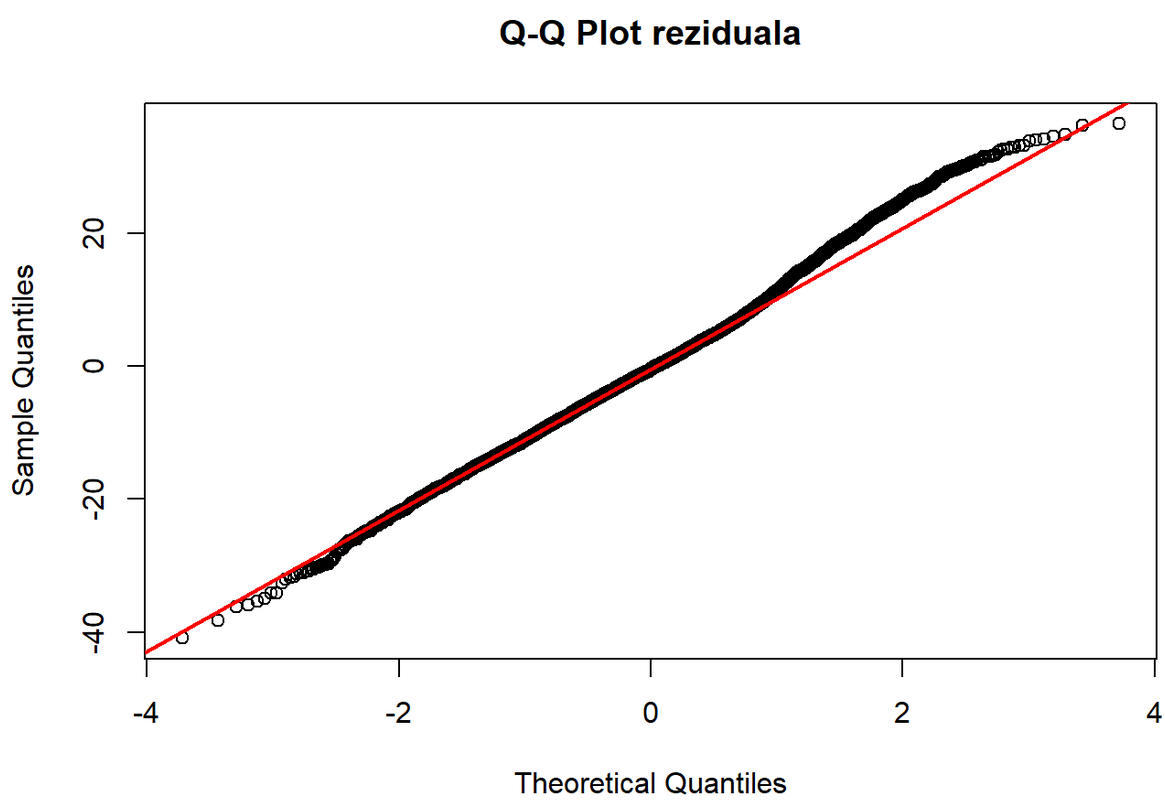
Razlika između R kvadrat i R-a je vrlo mala sto znaci da ukljucene varijable zaista pridonose objašnjenju varijance (nema penala!)

F statistika nam testira jesu li svi koeficijenti u modelu jednaki nuli. F = 2963 nam govori tj. p-value: < 2.2e-16 da je ta pretpostavka skoro pa nemoguca - odbacujemo H0.

```
residuals <- residuals(fit.multi)

hist(residuals, breaks = 30, col = "lightblue", main = "Glavni histogram reziduala naseg
modela",
     xlab = "Reziduali")
```




```
# uocavamo normalnu distribuciju reziduala naseg modela sto je jako bitno za validnost t
-testova

qqnorm(residuals, main = "Q-Q Plot reziduala")
qqline(residuals, col = "red", lwd = 2)
```



Sada

cemo na primjeru podataka jedne anonimne osobe predvidjeti njene godine. Koristimo 95% interval povjerenja

```
#diab_red

#names(diab_red)

#diab_red

new_data <- data.frame(BMI = 25,`Waist Circumference` = 85, `Blood Pressure` = 80, `Chol
esterol Levels` = 22)



new_data
```

```
##   BMI Waist.Circumference Blood.Pressure Cholesterol.Levels
## 1  25                  85             80                 22
```

```
colnames(new_data)[colnames(new_data) == "Blood.Pressure"] <- "Blood Pressure"
colnames(new_data)[colnames(new_data) == "Cholesterol.Levels"] <- "Cholesterol Levels"
colnames(new_data)[colnames(new_data) == "Waist.Circumference"] <- "Waist Circumference"


new_data
```

```
##   BMI Waist Circumference Blood Pressure Cholesterol Levels
## 1  25                  85             80                 22
```

```
pred <- predict(fit.multi, newdata = new_data, interval = "confidence", level = 0.95)

pred
```

```
##        fit      lwr      upr
## 1 35.75319 30.95275 40.55364
```

U ovom slučaju, predviđena starost je 36.25 godina. Donja granica 95% intervala poverenja: 31.47 godina. Gornja granica 95% intervala poverenja: 41.03 godina.

Zaključak:

Nas 95% interval povjerenja je [31.47,41.03]

2. Projektno pitanje

“Postoje li razlike u oblicima bolesti među pušačima i nepušačima?”

Koristimo χ^2 test nezavisnosti kako bismo ispitali jesu li dvije kategorijske varijable međusobno povezane. U ovom slučaju želimo utvrditi postoji li statistički značajna povezanost između vrste bolesti (Target) i statusa pušenja (Smoking Status).

Osnova χ^2 testa je nulta hipoteza (H_0) koja pretpostavlja da su varijable nezavisne, što znači da promjene u jednoj varijabli (npr. status pušenja) ne utječu na promjene u drugoj varijabli (npr. vrsta bolesti).

Hipoteza HO: Ne postoji značajna povezanost između rezultata po oblicima bolesti (Target) i statusa pušenja (Smoking Status) - nezavisne varijable. Hipoteza H1: Postoji značajna povezanost između rezultata po oblicima bolesti (Target) i statusa pušenja (Smoking Status) - zavisne varijable.

```
cat("Različite vrste bolesti (Target):", paste(unique(as.character(diabetes$Target)), co
llapse = ", "), "\n")
```

```
## Različite vrste bolesti (Target): Steroid-Induced Diabetes, Neonatal Diabetes Mellitu
s (NDM), Prediabetic, Type 1 Diabetes, Wolfram Syndrome, LADA, Type 2 Diabetes, Wolcott-
Rallison Syndrome, Secondary Diabetes, Type 3c Diabetes (Pancreatogenic Diabetes), Gesta
tional Diabetes, Cystic Fibrosis-Related Diabetes (CFRD), MODY
```

```
cat("\n")
```

```
cat("Kategorije pušačkog statusa (Smoking Status):", paste(unique(as.character(diabetes
$`Smoking Status`)), collapse = ", "), "\n")
```

```
## Kategorije pušačkog statusa (Smoking Status): Smoker, Non-Smoker
```

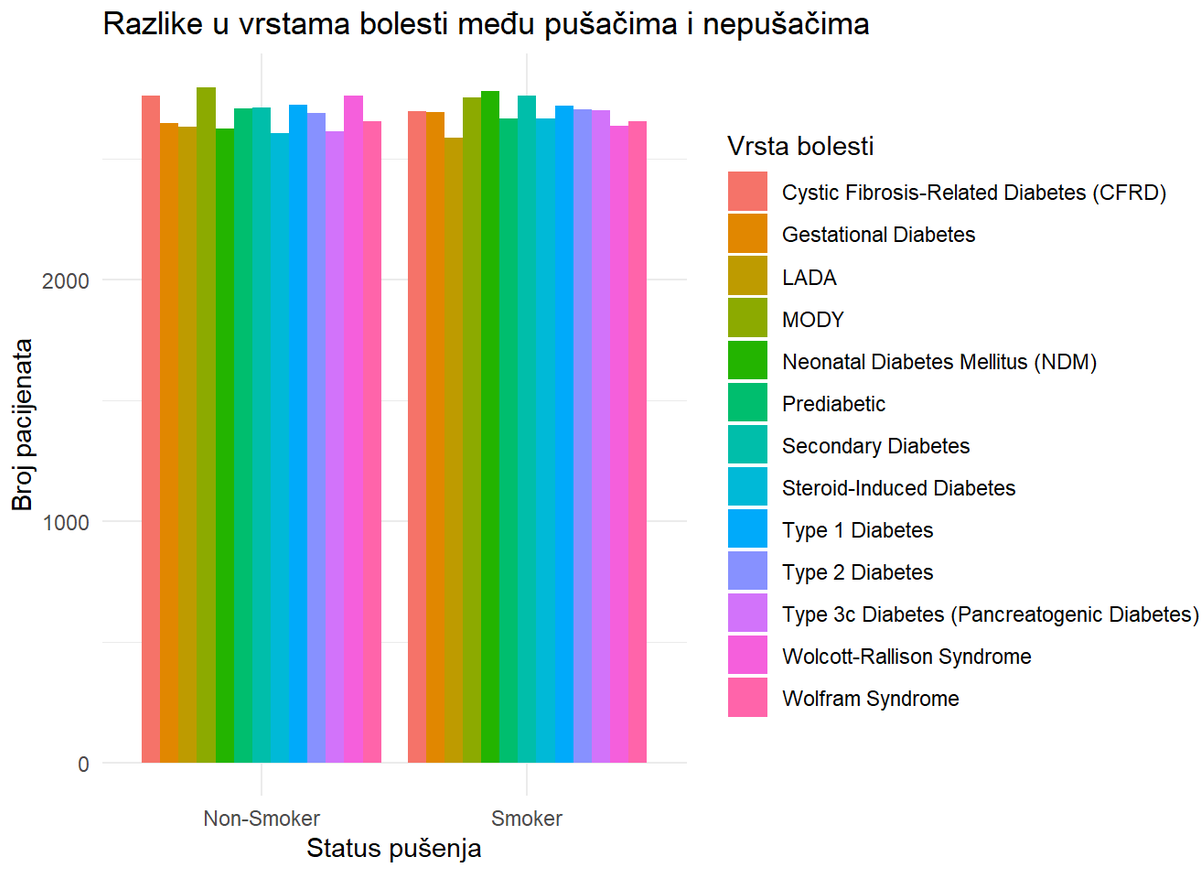
```
cat("\n")
```

```
target_smoking_table <- table(diabetes$Target, diabetes$`Smoking Status`)
kable(target_smoking_table, caption = "Dobivena frekvencija vrsta bolesti prema statusu
pušenja")
```

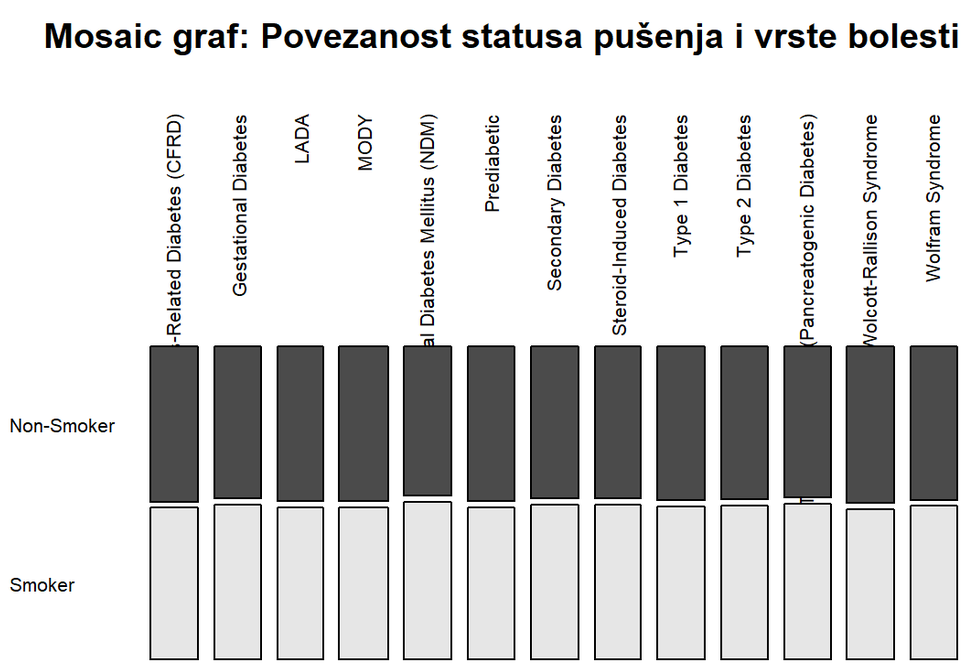
Dobivena frekvencija vrsta bolesti prema statusu pušenja

	Non-Smoker	Smoker
Cystic Fibrosis-Related Diabetes (CFRD)	2765	2699
Gestational Diabetes	2650	2694
LADA	2635	2588
MODY	2799	2754
Neonatal Diabetes Mellitus (NDM)	2626	2782
Prediabetic	2709	2667
Secondary Diabetes	2714	2765
Steroid-Induced Diabetes	2607	2668
Type 1 Diabetes	2725	2721
Type 2 Diabetes	2691	2706
Type 3c Diabetes (Pancreatogenic Diabetes)	2616	2704
Wolcott-Rallison Syndrome	2762	2638
Wolfram Syndrome	2656	2659

```
#Vizualizacija bar plot
ggplot(diabetes, aes(x = `Smoking Status`, fill = Target)) +
  geom_bar(position = "dodge") +
  labs(title = "Razlike u vrstama bolesti među pušačima i nepušačima",
       x = "Status pušenja",
       y = "Broj pacijenata",
       fill = "Vrsta bolesti") +
  theme_minimal()
```




```
# Vizualizacija podataka – Mosaic plot
mosaicplot(target_smoking_table, main = "Mosaic graf: Povezanost statusa pušenja i vrste
bolesti", color = TRUE, las = 2)
```




```
smoking_vs_target <- chisq.test(table(diabetes$Target, diabetes$`Smoking Status`))

cat("Rezultati χ² testa:\n")
```

```
## Rezultati χ² testa:
```

```
print(smoking_vs_target)
```

```
##
##  Pearson's Chi-squared test
##
## data:  table(diabetes$Target, diabetes$`Smoking Status`)
## X-squared = 12.189, df = 12, p-value = 0.4306
```

Iz grafičkih prikaza ne vidimo velike razlike, tj. frekvencije su otprilike podjednake za svaku vrstu bolesti odnosno status pušenja.

```
contingency_table <- table(diabetes$Target, diabetes$`Smoking Status`)
# Chi-squared test

chi_squared_test <- chisq.test(contingency_table)
expected_frequencies <- chi_squared_test$expected

# Prikaz očekivanih frekvencija
#kable(expected_frequencies, caption = "Očekivane frekvencije vrsta bolesti prema status
u pušenja")

# Ispis frekvencijske tablice s očekivanim frekvencijama
smoking_levels <- levels(diabetes$`Smoking Status`)
target_levels <- levels(diabetes$Target)
# Create a matrix with the combined values
combined_table <- data.frame(
  Smoking_Status = rep(smoking_levels, each = length(target_levels)),
  Target = rep(target_levels, times = length(smoking_levels)),
  Dobivene_Frekvencije = as.vector(contingency_table),
  Ocekivane_Frekvencije = as.vector(expected_frequencies)
)
# Prikaz tablice sa očekivanim i dobivenim frekvencijama
kable(combined_table, caption = "Stvarne i očekivane frekvencije za svaku kombinaciju va
rijabli")
```

Stvarne i očekivane frekvencije za svaku kombinaciju varijabli

Smoking_Status	Target	Dobivene_Frekvencije	Ocekivane_Frekvencije
Non-Smoker	Cystic Fibrosis-Related Diabetes (CFRD)	2765	2728.487
Non-Smoker	Gestational Diabetes	2650	2668.565
Non-Smoker	LADA	2635	2608.142
Non-Smoker	MODY	2799	2772.930
Non-Smoker	Neonatal Diabetes Mellitus (NDM)	2626	2700.523
Non-Smoker	Prediabetic	2709	2684.544
Non-Smoker	Secondary Diabetes	2714	2735.978
Non-Smoker	Steroid-Induced Diabetes	2607	2634.109
Non-Smoker	Type 1 Diabetes	2725	2719.499
Non-Smoker	Type 2 Diabetes	2691	2695.030
Non-Smoker	Type 3c Diabetes (Pancreatogenic Diabetes)	2616	2656.580
Non-Smoker	Wolcott-Rallison Syndrome	2762	2696.529
Non-Smoker	Wolfram Syndrome	2656	2654.083
Smoker	Cystic Fibrosis-Related Diabetes (CFRD)	2699	2735.513
Smoker	Gestational Diabetes	2694	2675.435
Smoker	LADA	2588	2614.858
Smoker	MODY	2754	2780.070

Smoking_Status	Target	Dobivene_Frekvencije	Ocekivane_Frekvencije
Smoker	Neonatal Diabetes Mellitus (NDM)	2782	2707.477
Smoker	Prediabetic	2667	2691.456
Smoker	Secondary Diabetes	2765	2743.022
Smoker	Steroid-Induced Diabetes	2668	2640.891
Smoker	Type 1 Diabetes	2721	2726.501
Smoker	Type 2 Diabetes	2706	2701.970
Smoker	Type 3c Diabetes (Pancreatogenic Diabetes)	2704	2663.420
Smoker	Wolcott-Rallison Syndrome	2638	2703.471
Smoker	Wolfram Syndrome	2659	2660.917

```
df <- (nrow(contingency_table) - 1) * (ncol(contingency_table) - 1)

# Za stupanj slobode df = (broj redaka - 1) * (broj stupaca - 1)
# Kritična vrijednost testa
alpha <- 0.05
critical_value <- qchisq(1 - alpha, df)

cat("Stupanj slobode:" ,df, "\n")
```

```
## Stupanj slobode: 12
```

```
cat("Kritična vrijednost χ² testa:", critical_value, "\n")
```

```
## Kritična vrijednost χ² testa: 21.02607
```

```
cat("Vrijednost χ² testa:", chi_squared_test$statistic, "\n")
```

```
## Vrijednost χ² testa: 12.18902
```

```
cat("P-vrijednost (p-value):", chi_squared_test$p.value, "\n")
```

```
## P-vrijednost (p-value): 0.4306211
```

Zaključak:

Iz dobivenih rezultata, vidimo da je p-vrijednost veća od 0.05 te ne možemo odbaciti nultu hipotezu(H0) tj. Ne postoji značajna razlika među oblicima bolesti i statusa pušenja. Nema dovoljno dokaza da zaključimo da su Target(Vrsta bolesti) i Smoking Status(Status pušenja) značajno povezani na razini sigurnosti 95%.

3. Projektno pitanje

U ovom dijelu razmatram postoji li razlika u razini zdravlja gušteraće među oblicima dijabetesa pacijenta?

Target je kategorijski tip podataka, dok je razina zdravlja gušterače numerički tip podatka.

Koristimo ANOVA test s obzirom da uspoređujemo srednje vrijednosti više skupina jedne kategorijske varijable s numeričkom varijablom. ANOVA test omogućuje nam analizu utjecaja različitih čimbenika na jednu varijablu, što je važno za utvrđivanje postoje li statistički značajne razlike između grupa.

Hipoteza H0: Ne postoji značajna razlika u razini zdravlja gušterače po oblicima dijabetesa Hipoteza H1: Postoji značajna razlika u razini zdravlja gušterače po oblicima dijabetesa

```
#kopiramo podatke kako ne bi promijenili pravi dataset
diabetes_copy2 = data.frame(diabetes)

#names(diabetes_copy2)

#factor(diabetes_copy2$`Pancreatic.Health`)
#factor(diabetes_copy2$Target)

#diabetes$`Target`
class(diabetes$`Target`)
```

```
## [1] "factor"
```

```
#diabetes$`Pancreatic Health`
class(diabetes$`Pancreatic Health`)
```

```
## [1] "numeric"
```

ANOVA test pretpostavlja homogenost i normalnost pa moramo prvo provjeriti naše podatke kako bi mogli krenuti sa ANOVA testom. Provjeravamo normalnost podataka za svaku pojedinu grupu dijabetsa pomoću Lillieforsovom inačicom Kolmogorov-Smirnov testa

```
require(nortest)
```

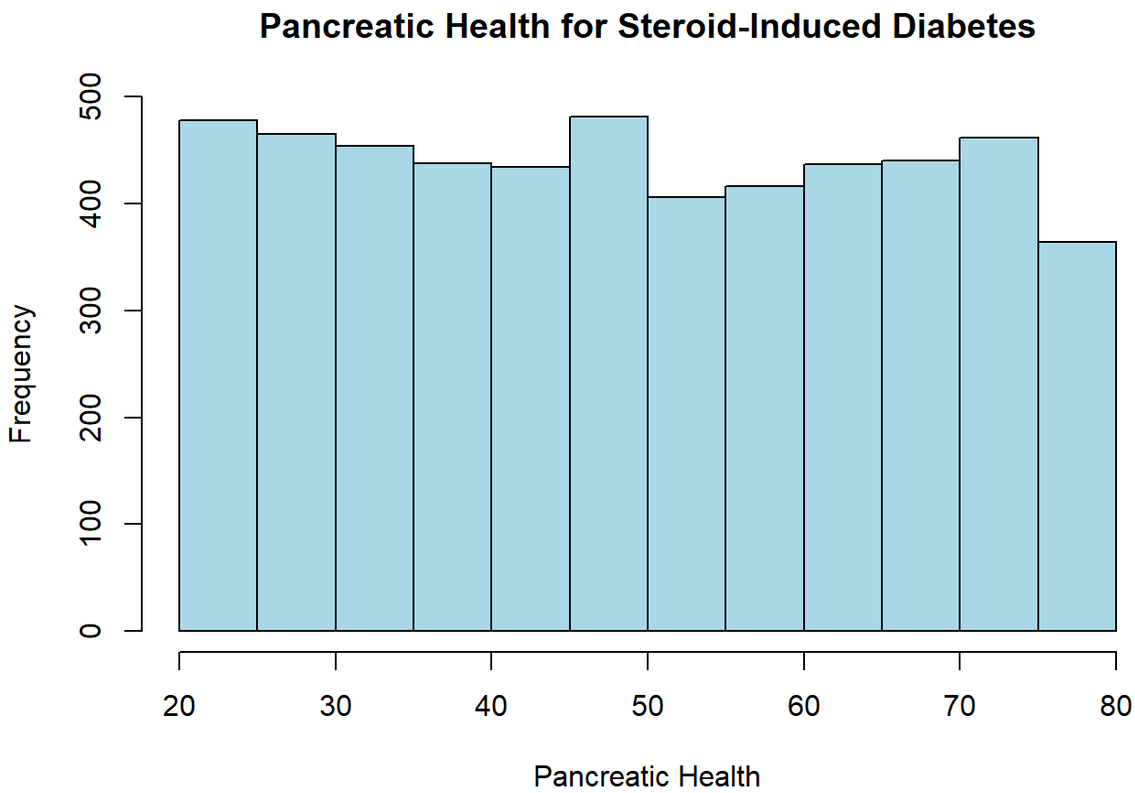
```
## Loading required package: nortest
```

```
lillie.test(diabetes_copy2$Pancreatic.Health)
```

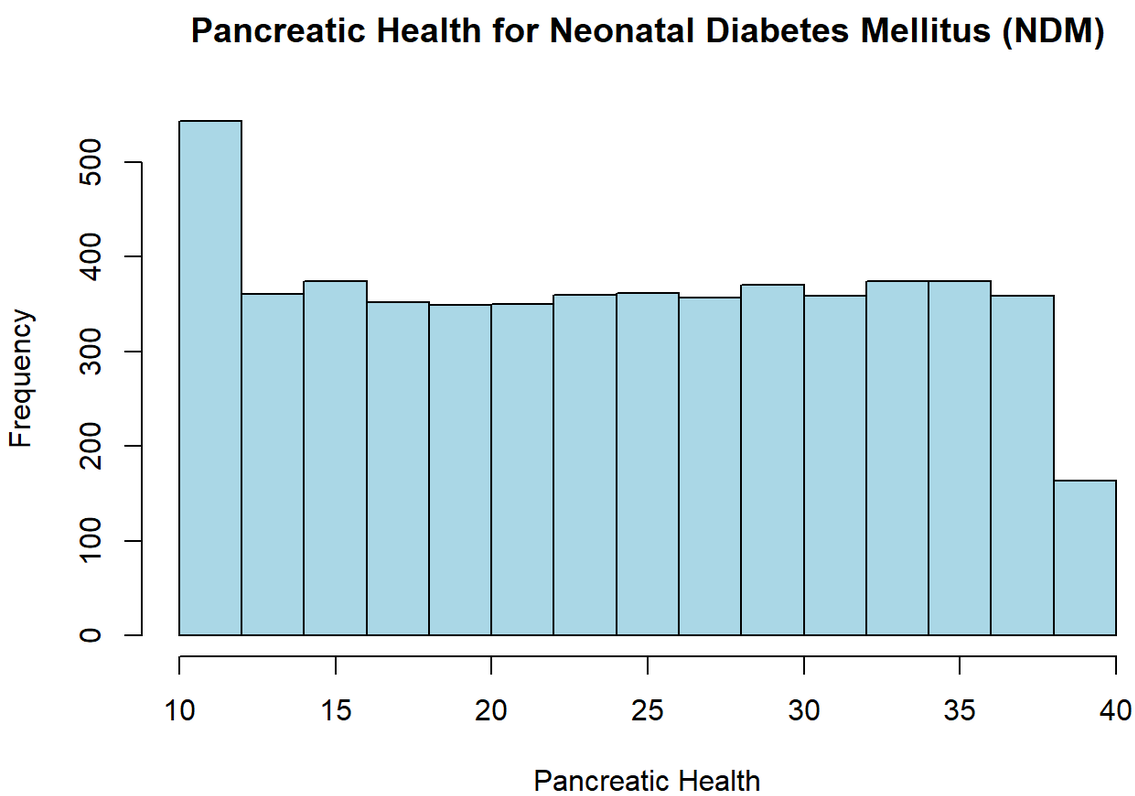
```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  diabetes_copy2$Pancreatic.Health
## D = 0.068414, p-value < 2.2e-16
```

```
# Lista tipova dijabetesa
diabetes_types <- unique(diabetes_copy2$Target)

# Kreiranje histograma za svaki tip dijabetesa
for (diabetes_type in diabetes_types) {
  # Filtriranje podataka za određeni tip dijabetesa
  data_subset <- diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target == diabetes_typ
e]

  #lillie-test
  lillie_result <- lillie.test(data_subset)
  print(paste("Lillie test result for", diabetes_type, ":", lillie_result$p.value))
    # Kreiranje histograma
  hist(data_subset,
       main = paste("Pancreatic Health for", diabetes_type),  # Naslov sa tipom dijabete
sa
       xlab = "Pancreatic Health",  # Oznaka x-ose
       col = "lightblue",  # Boja histograma (opcionalno)
       border = "black")  # Boja ivica (opcionalno)
}
```

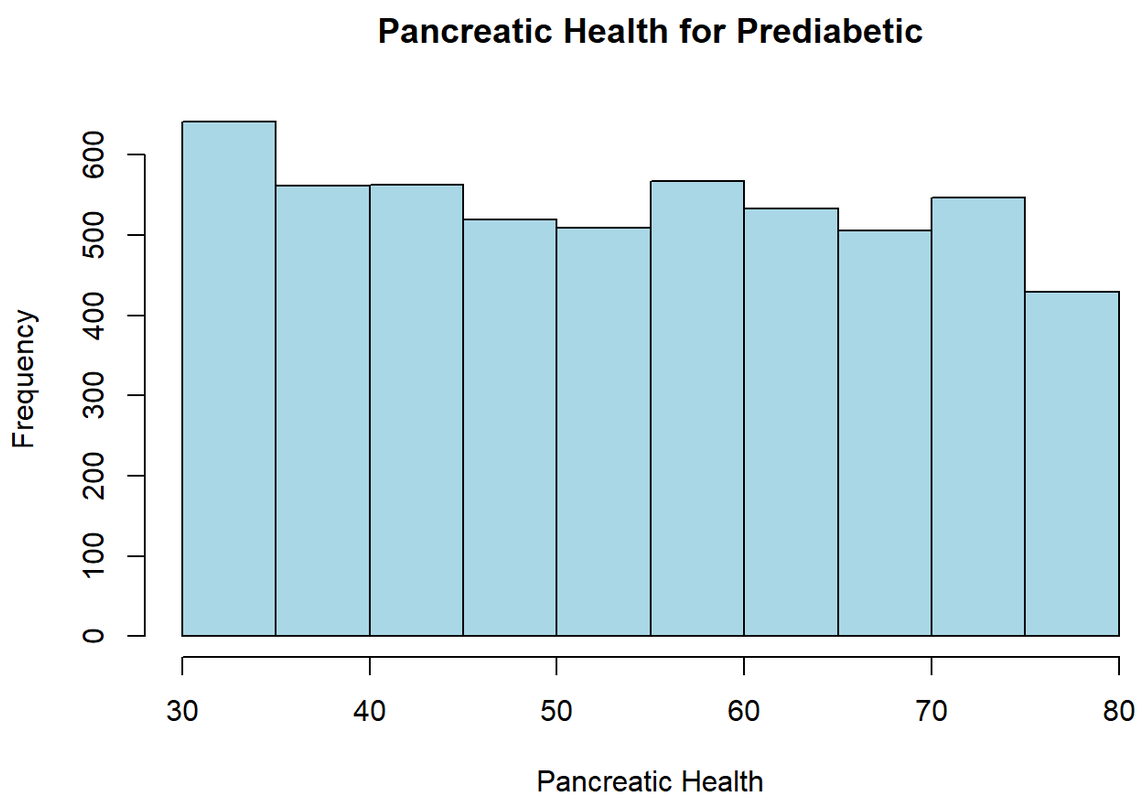
```
## [1] "Lillie test result for Steroid-Induced Diabetes : 1.0461707480236e-73"
```



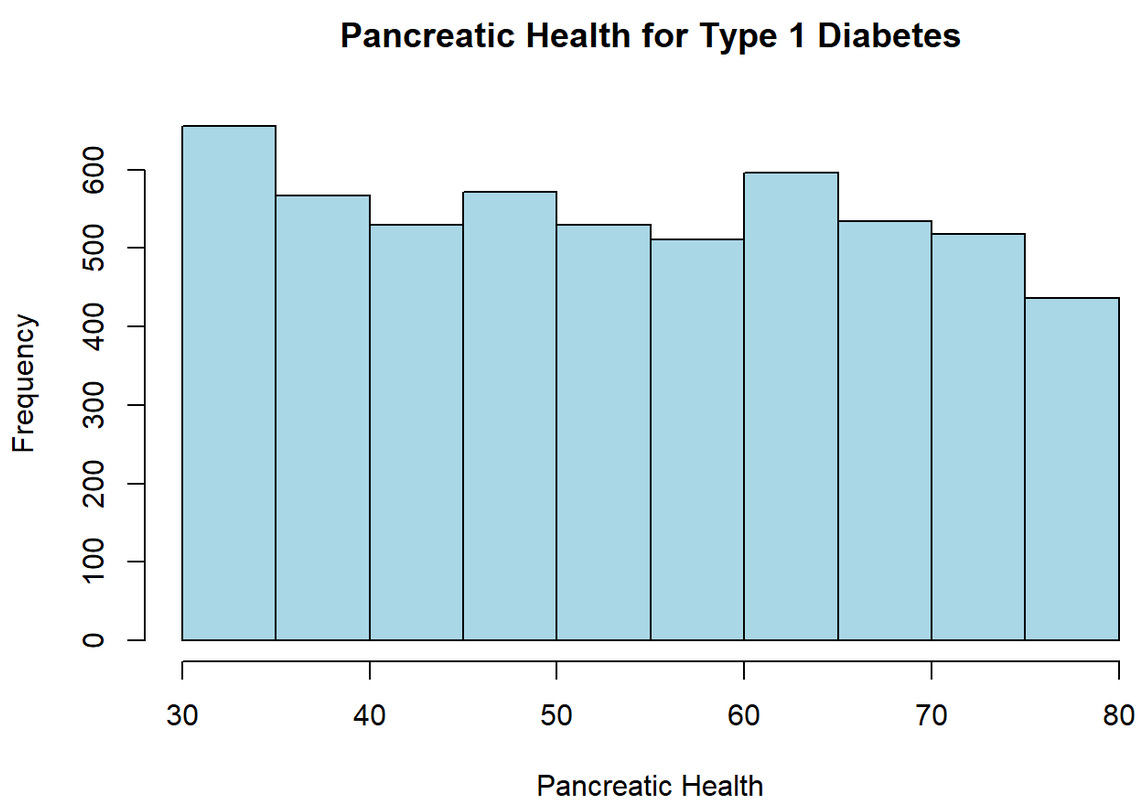

```
## [1] "Lillie test result for Neonatal Diabetes Mellitus (NDM) : 1.00768949161913e-86"
```



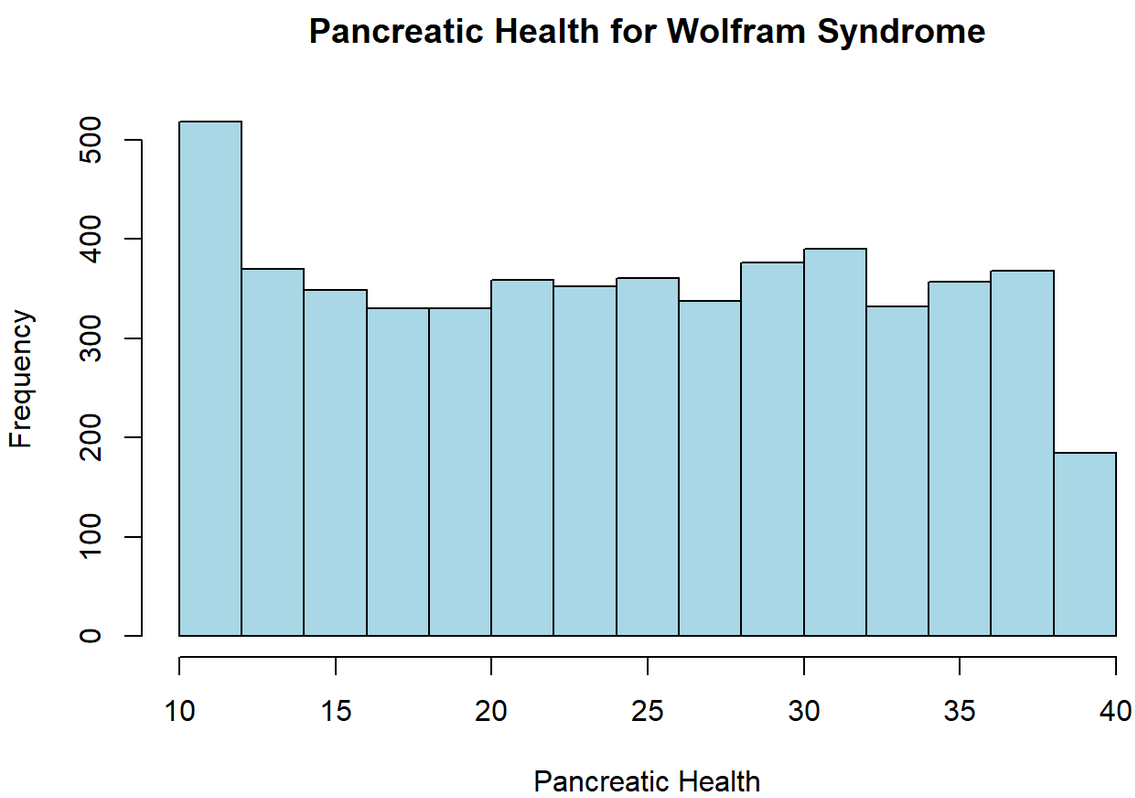

```
## [1] "Lillie test result for Prediabetic : 1.53545208640135e-73"
```



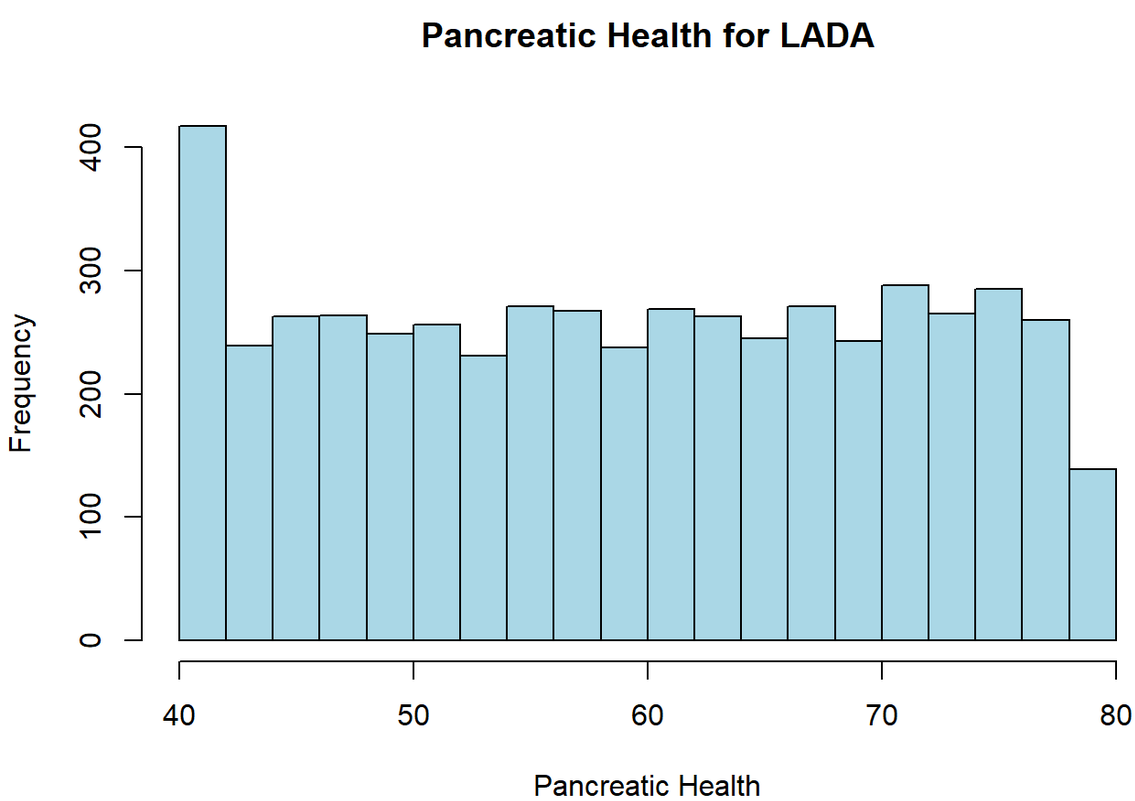

```
## [1] "Lillie test result for Type 1 Diabetes : 4.26370855086212e-72"
```



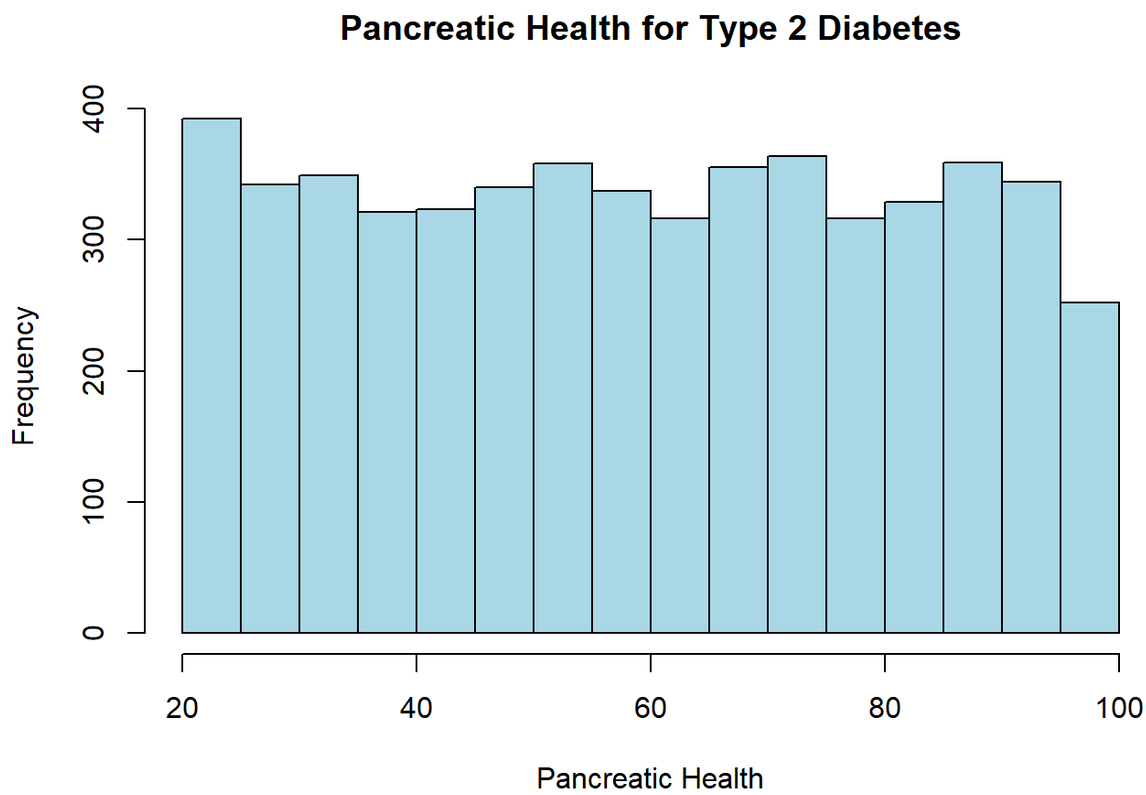

```
## [1] "Lillie test result for Wolfram Syndrome : 4.54340602830596e-81"
```



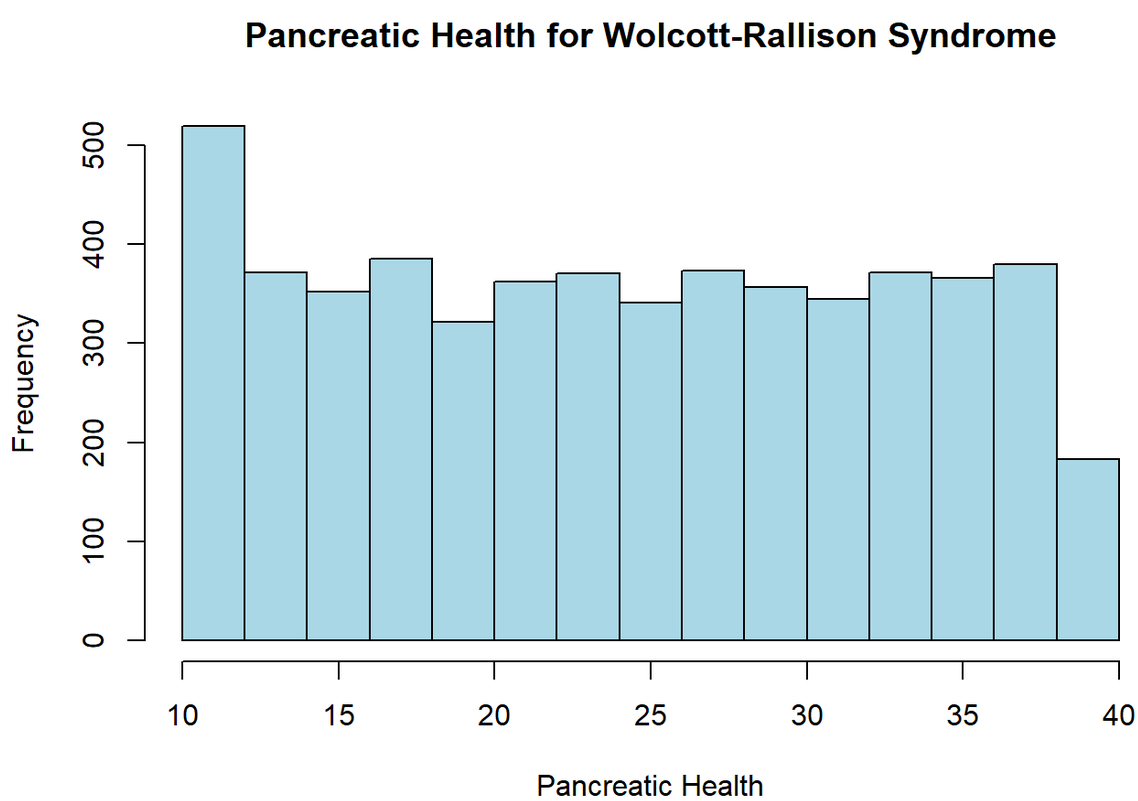

```
## [1] "Lillie test result for LADA : 9.58723988923272e-74"
```



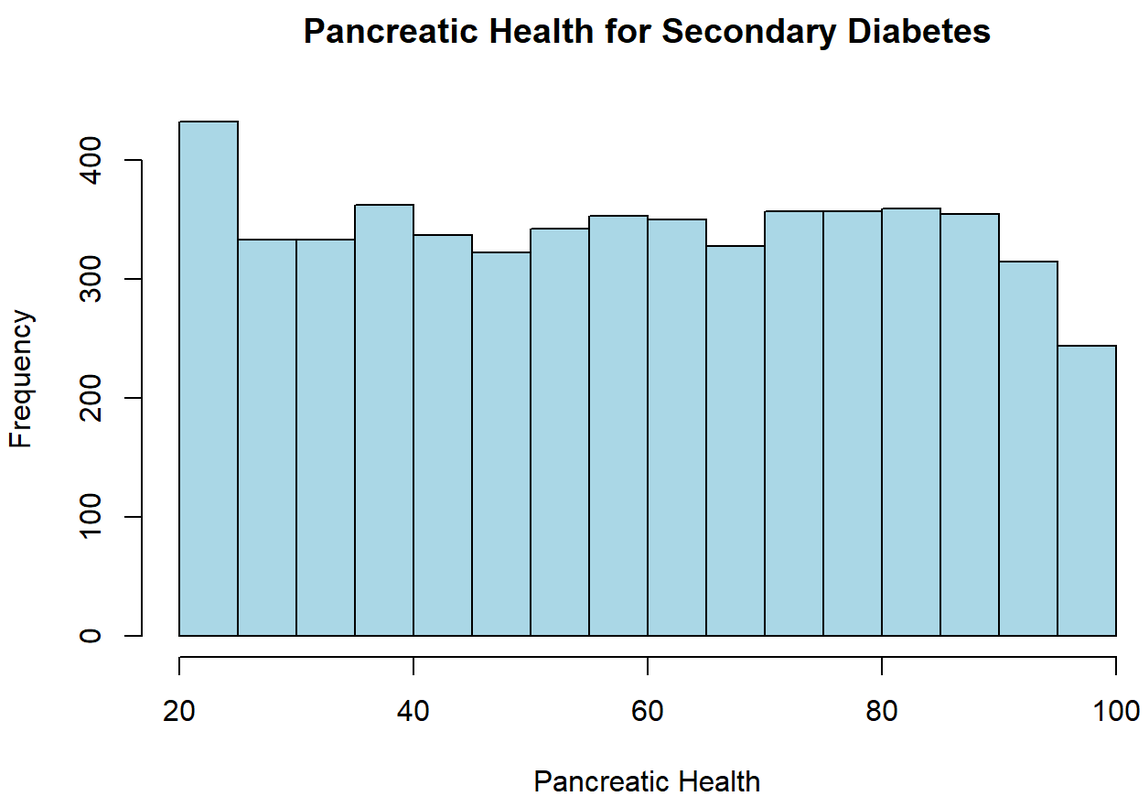

```
## [1] "Lillie test result for Type 2 Diabetes : 7.5148662012579e-59"
```



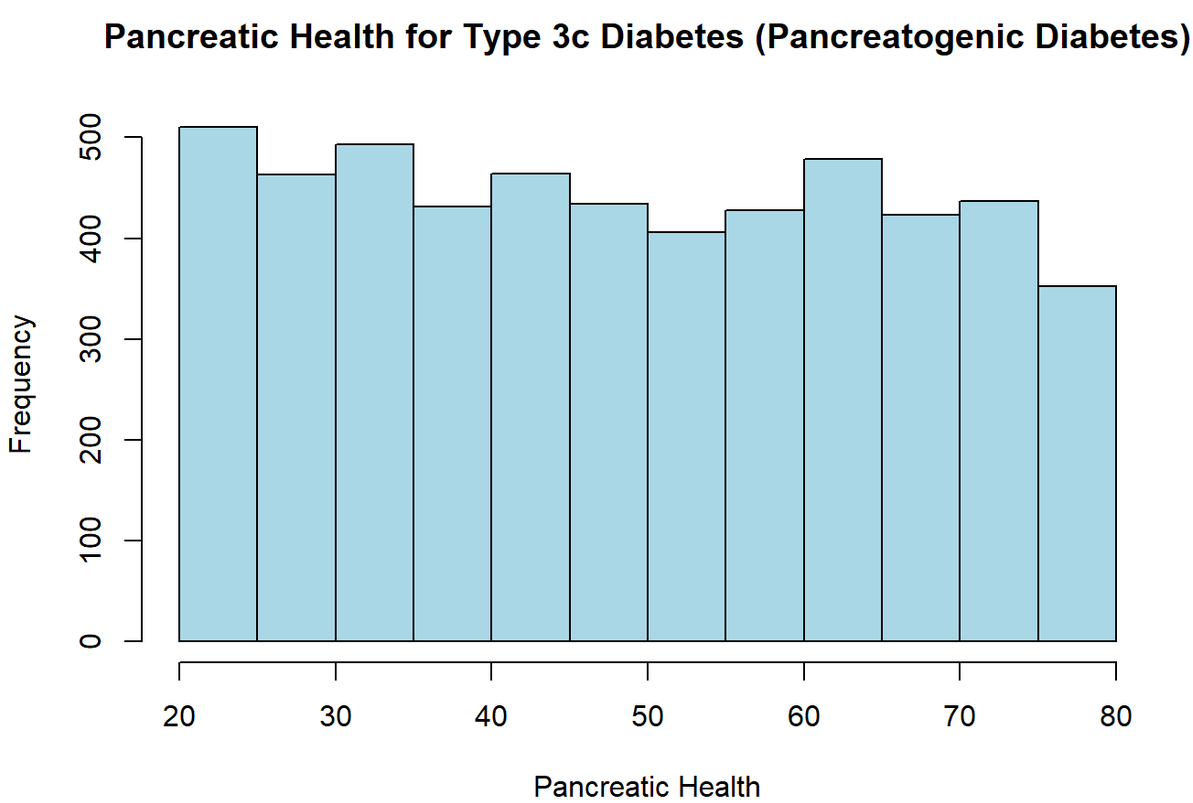

```
## [1] "Lillie test result for Wolcott-Rallison Syndrome : 2.99834984600884e-88"
```



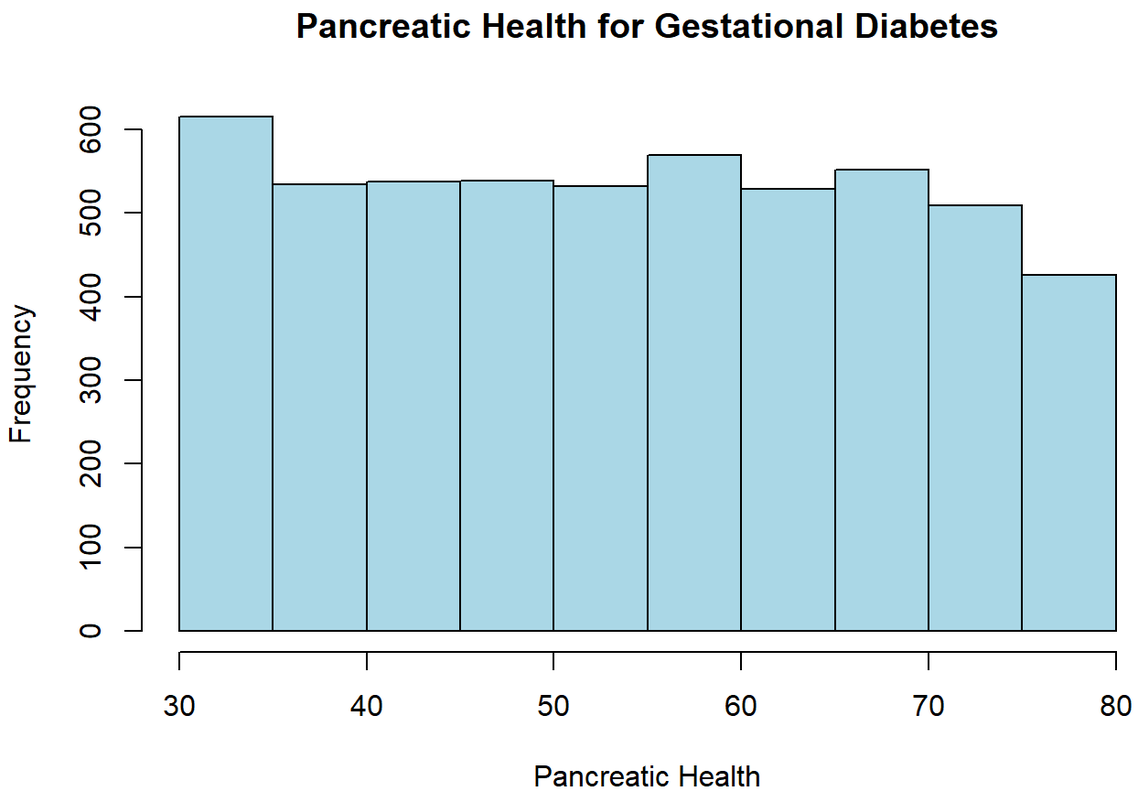

```
## [1] "Lillie test result for Secondary Diabetes : 1.60476108964642e-66"
```



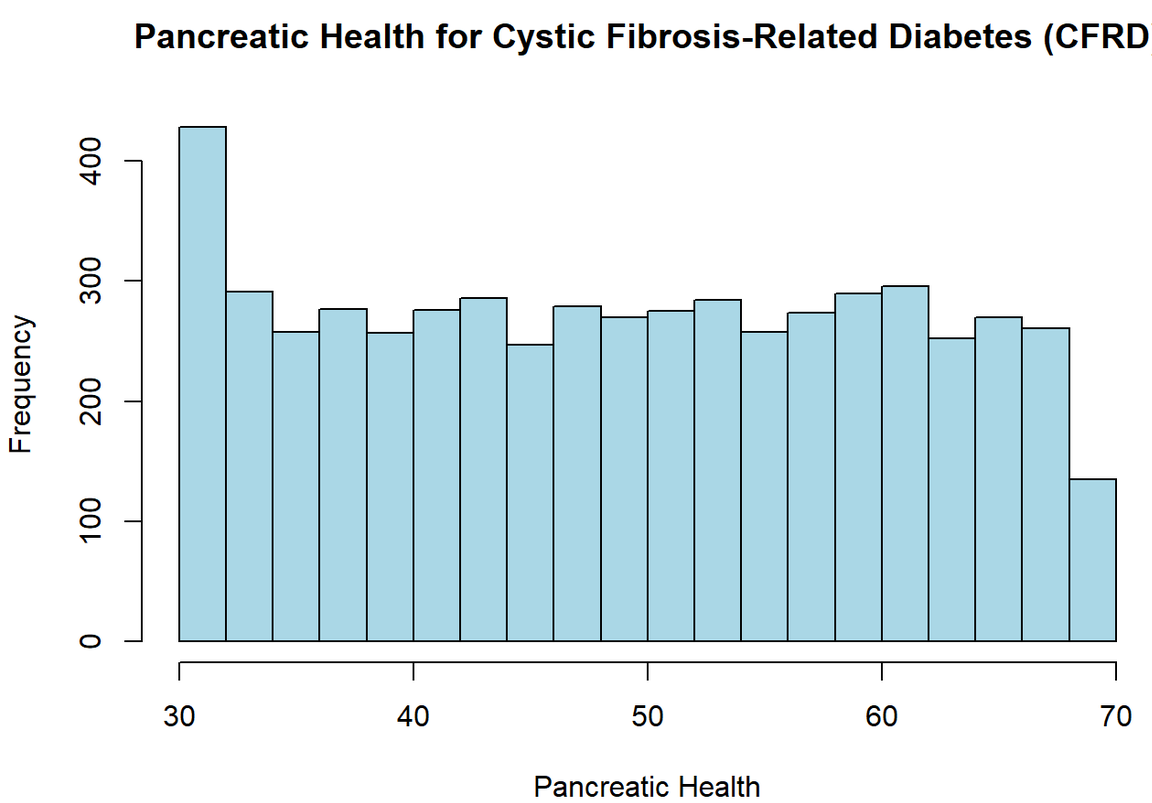

```
## [1] "Lillie test result for Type 3c Diabetes (Pancreatogenic Diabetes) : 7.2501211911
8581e-73"
```



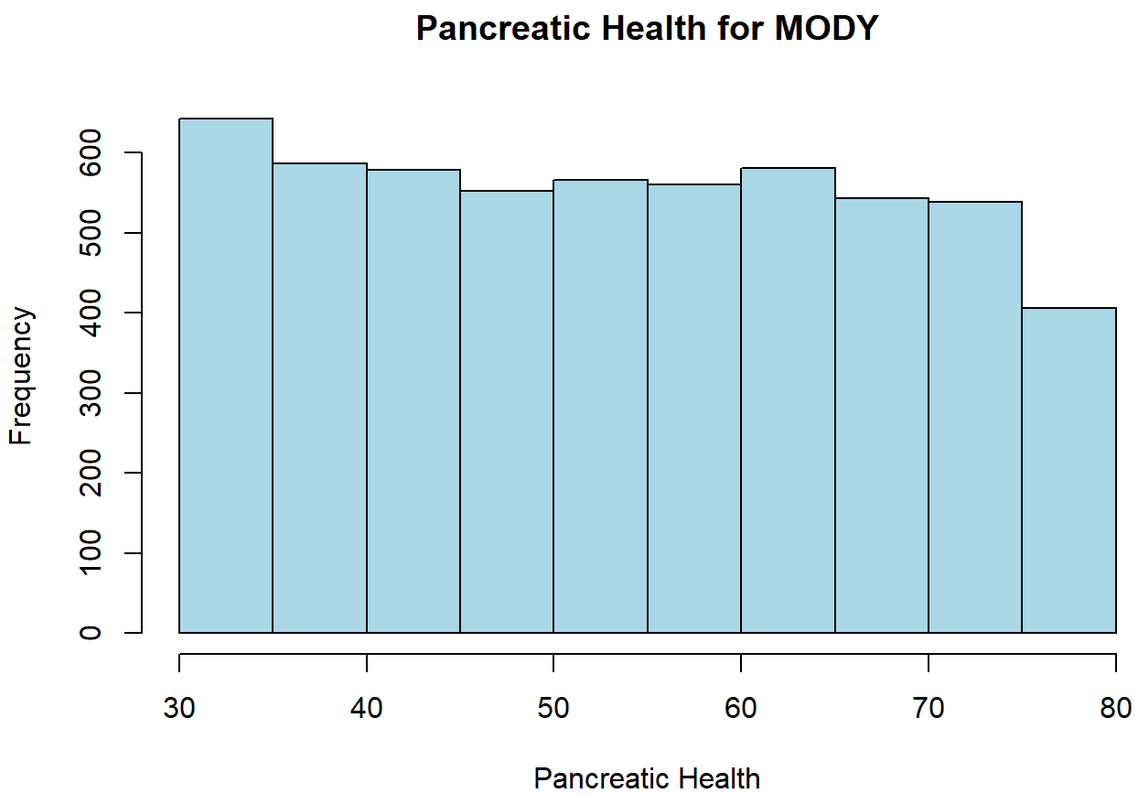

```
## [1] "Lillie test result for Gestational Diabetes : 6.22617556022835e-68"
```




```
## [1] "Lillie test result for Cystic Fibrosis-Related Diabetes (CFRD) : 1.0295819880421
9e-75"
```




```
## [1] "Lillie test result for MODY : 2.21551404977202e-74"
```




```
#testiranje homogenosti varijance uzorka
bartlett.test(diabetes_copy2$Pancreatic.Health ~ diabetes_copy2$Target)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  diabetes_copy2$Pancreatic.Health by diabetes_copy2$Target
## Bartlett's K-squared = 14096, df = 12, p-value < 2.2e-16
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Wolfram Syndrome'])
```

```
## [1] 75.46892
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Gestational Diabetes'])
```

```
## [1] 205.5873
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='LADA'])
```

```
## [1] 135.6495
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='MODY'])
```

```
## [1] 203.6217
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Neonatal Diabetes Mellitus
(NDM)'])
```

```
## [1] 75.23981
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Prediabetic'])
```

```
## [1] 209.5374
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Secondary Diabetes'])
```

```
## [1] 529.6986
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Steroid-Induced Diabetes'])
```

```
## [1] 299.7283
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Type 1 Diabetes'])
```

```
## [1] 206.4028
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Type 2 Diabetes'])
```

```
## [1] 531.1273
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Type 3c Diabetes (Pancreato
genic Diabetes)'])
```

```
## [1] 299.9355
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Wolcott-Rallison Syndrom
e'])
```

```
## [1] 75.20566
```

```
var(diabetes_copy2$Pancreatic.Health[diabetes_copy2$Target=='Wolfram Syndrome'])
```

```
## [1] 75.46892
```

Vidimo kako podatci nisu normalno distribuirano. Također testirali smo i homogenost varijance uzorka. *Podatci ne zadovoljavaju uvjet homogenosti niti su normalno distribuirani* (jer su dobivene p-vrijednosti manje od 0.05). Iz toga zaključujemo kako ne možemo koristiti ANOVA test.

Zaključujemo kako trebamo koristiti neparametarski test - Kruskal - Wallis. Taj je test neparametarski ekvivalent jednosmjernog ANOVA testa.

```
#kopiramo podatke kako ne bi promijenili pravi dataset
diabetes_copy3 = data.frame(diabetes)
```

Histogram zdravlja gušterače

```
# 1. Aritmetička sredina
mean_pancreas <- mean(diabetes_copy3$`Pancreatic.Health`)
mean_pancreas
```

```
## [1] 47.56424
```

```
# 2. Podrezana (trimmed) sredina (npr. 20%)
mean_trimmed_pancreas <- mean(diabetes_copy3$`Pancreatic.Health`, trim = 0.2)
mean_trimmed_pancreas
```

```
## [1] 46.94283
```

```
# 3. Medijan
median_pancreas <- median(diabetes_copy3$`Pancreatic.Health`)
median_pancreas
```

```
## [1] 46
```

```
# 4. Kvartili (Q1 i Q3)
quantiles_pancreas <- quantile(diabetes_copy3$`Pancreatic.Health`, probs = c(0.25, 0.7
5))
quantiles_pancreas
```

```
## 25% 75%
##  32  64
```

```
# 5. Mod (najčešća vrijednost) – paket modeest
require(modeest)
```

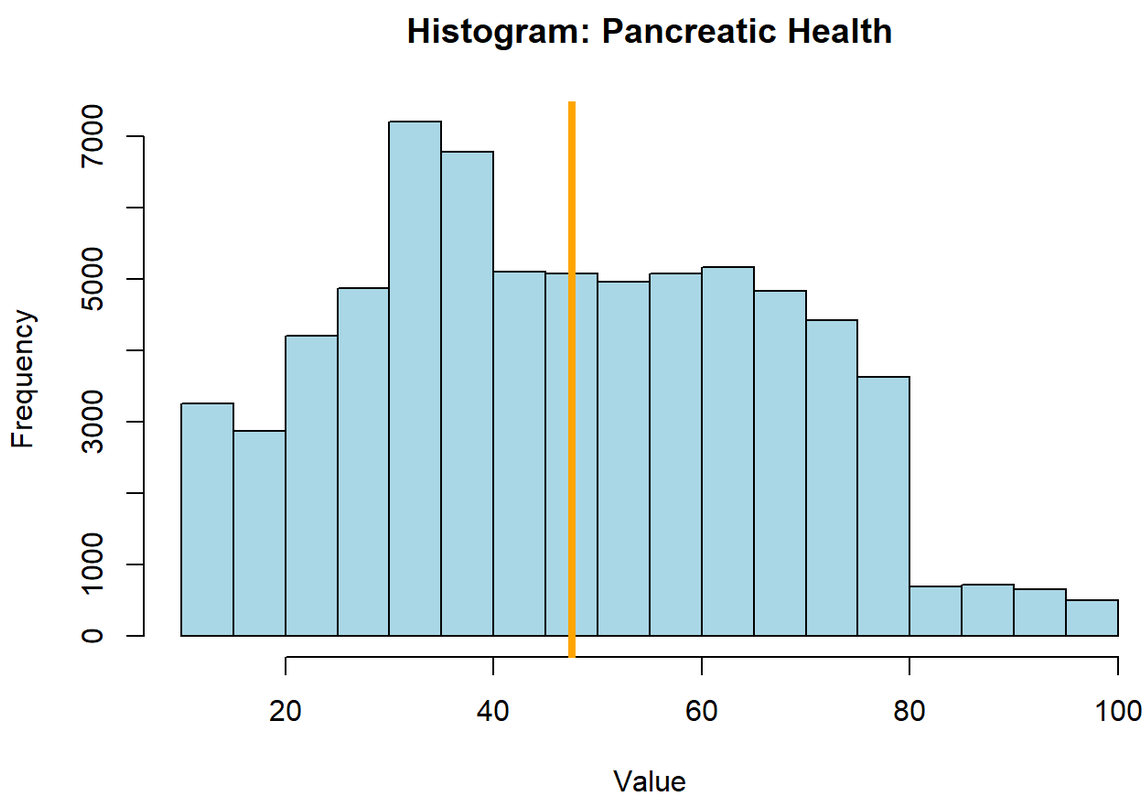
```
## Loading required package: modeest
```

```
## Warning: package 'modeest' was built under R version 4.4.2
```

```
mode_pancreas <- mfv(diabetes_copy3$`Pancreatic.Health`)
mode_pancreas
```

```
## [1] 37
```

```
# 6. Histogram
hist(diabetes_copy3$`Pancreatic.Health`,
     main = "Histogram: Pancreatic Health",
     xlab = "Value",
     ylab = "Frequency",
     col = "lightblue")

# Dodajemo okomitu liniju u boji narančaste, debljina 4
abline(v = mean(diabetes_copy3$`Pancreatic.Health`),
       col = "orange", lwd = 4)
```

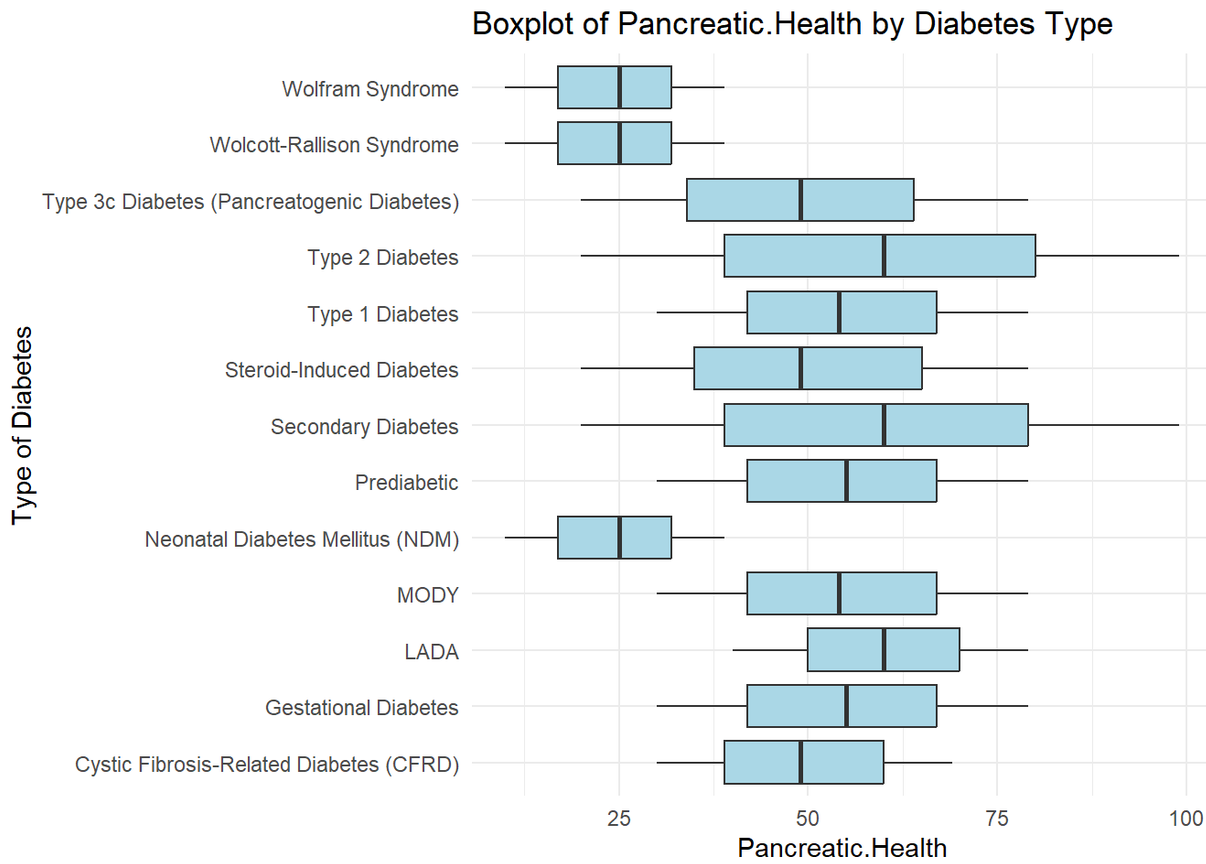



```
# 7. Summary (min, Q1, median, mean, Q3, max)
summary(diabetes_copy3$`Pancreatic.Health`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   10.00   32.00   46.00   47.56   64.00   99.00
```

Napraviti ćemo box plot prikaz svih podataka.

```
# Boxplot: Pancreatic.Health ~ Target
ggplot(data = diabetes_copy3, aes(x = Target, y = Pancreatic.Health)) +
  geom_boxplot(
    fill = "lightblue",
    outlier.colour = "red"
  ) +

  labs(
    title = "Boxplot of Pancreatic.Health by Diabetes Type",
    x = "Type of Diabetes",
    y = "Pancreatic.Health"
  ) +
  coord_flip() +
  theme_minimal()
```



Grafički nam prikaz ukazuje na to da vrsta dijabetesa ima veze sa zdravljem gušteraće, no to ne možemo potvrditi bez da napravimo Kruskal-Wallis test

Izrada Kruskal-Wallis testa:

```
kruskal.test(Pancreatic.Health ~ Target, data = diabetes_copy3)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Pancreatic.Health by Target
## Kruskal-Wallis chi-squared = 30882, df = 12, p-value < 2.2e-16
```

Zaključak:

*Iz rezultata Kruskal-Wallis testa možemo odbaciti nultu hipotezu(H0), jer je p-vrijednost manja od 0.05 (dobivena p-vrijednost < 2.2e-16). Odnosno odbacujemo hipotezu da nema razlike u zdravlju gušterače prema oblicima dijabetesa, te se prihvaća alternativna hipoteza (H1) da postoji značajna razlika u zdravlju gušterače po oblicima dijabetesa. Odnosno barem jedan tip dijabetesa ima znatno drugačije razlike zdravlja gušteraće od drugih.

4. Projektno pitanje

“Utječe li razina fiziče aktivnosti na razinu inzulina u krvi pacijenta?”

Hipoteza H0: Ne postoji značajna razlika razine inzulina u krvi između grupa različite fizičke aktivnosti

Hhipoteza H1: Postoji razlika razine inzulina u krvi između grupa različite fizičke aktivnosti

```
#diabetes$`Physical Activity`
class(diabetes$`Physical Activity`)
```

```
## [1] "factor"
```

```
#diabetes$`Insulin Levels`
class(diabetes$`Insulin Levels`)
```

```
## [1] "numeric"
```

```
diabetes[c(1,6,9)]
```

```
## # A tibble: 70,000 × 3
##    Target                           `Insulin Levels` `Physical Activity`
##    <fct>                                       <dbl> <fct>
##  1 Steroid-Induced Diabetes                       40 High
##  2 Neonatal Diabetes Mellitus (NDM)               13 High
##  3 Prediabetic                                    27 High
##  4 Type 1 Diabetes                                 8 Low
##  5 Wolfram Syndrome                               17 High
##  6 LADA                                           17 Moderate
##  7 Type 2 Diabetes                                29 Moderate
##  8 Wolcott-Rallison Syndrome                      10 Low
##  9 Secondary Diabetes                             47 High
## 10 Secondary Diabetes                             21 Low
## # ℹ 69,990 more rows
```

Opis i pregled izdvojenih atributa

- Fizička aktivnost i razina inzulina

Fizička aktivnost

- kategorički je tip podatatka
- može biti vrijednosti: “Low”, “Moderate” ili “High”

```
#Mod
require(modeest)
#Najzastupljenija vrijednost fizičke aktivnosti vraća funkcija ispod
mfv(diabetes$`Physical Activity`)
```

```
## [1] Moderate
## Levels: High Low Moderate
```

```
#Provjera ima li nekavih rezulatta koji odstupaju od okvira
table(diabetes$`Physical Activity`)
```

```
##
##     High      Low Moderate
##    23225    23348    23427
```

Razina inzulina

- numerički je tip podatka

```
v = mean(diabetes$`Insulin Levels`)

# Podrezana aritmeticka sredina (20%)
mean(diabetes$`Insulin Levels`, trim=0.2)
```

```
## [1] 19.99071
```

```
median(diabetes$`Insulin Levels`)
```

```
## [1] 19
```

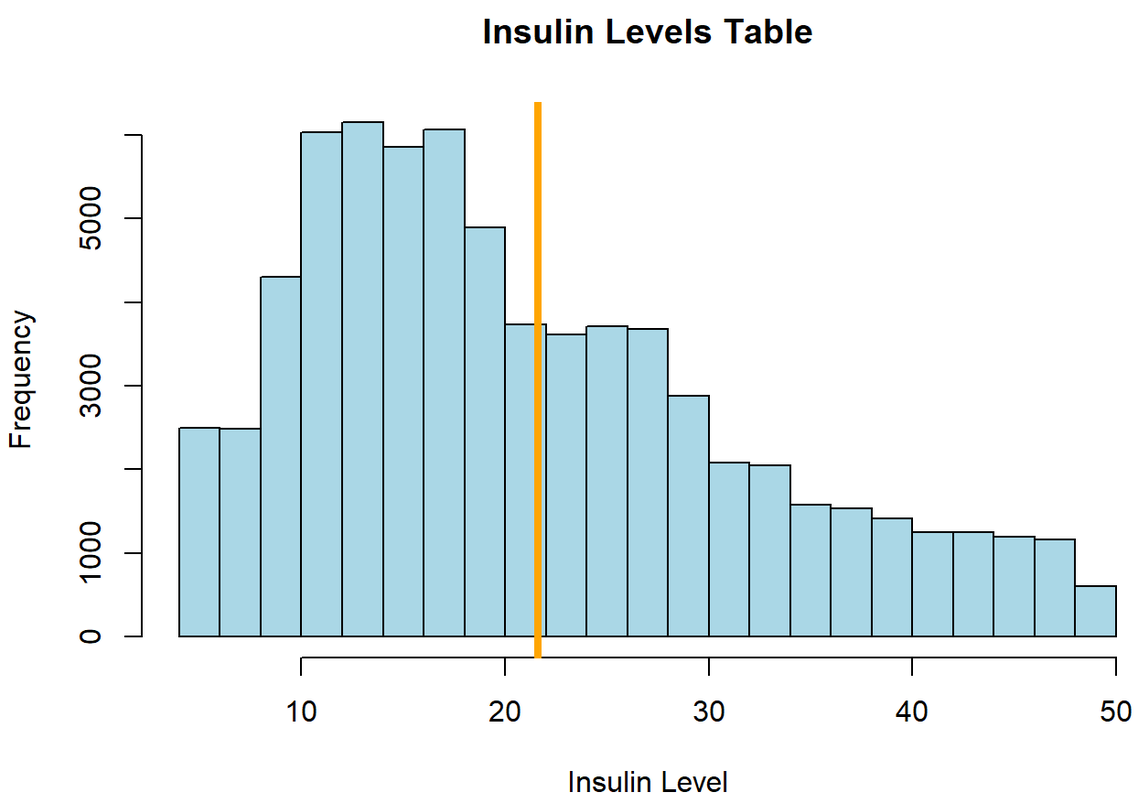
```
quantile(diabetes$`Insulin Levels`, probs = c(0.25,0.75))
```

```
## 25% 75%
##  13  28
```

```
#Mod
require(modeest)
mfv(diabetes$`Insulin Levels`)
```

```
## [1] 13
```

```
#Crtanje histograma koji prikazuje rapodjelu inzulina po vrijednosti i frekvenciji
h = hist(diabetes$`Insulin Levels`,
         #prob=TRUE,
         main="Insulin Levels Table",
         xlab="Insulin Level",
         ylab='Frequency',
         col="lightblue"
         )

abline(v = mean(diabetes$`Insulin Levels`), col = "orange", lwd = 4)
```



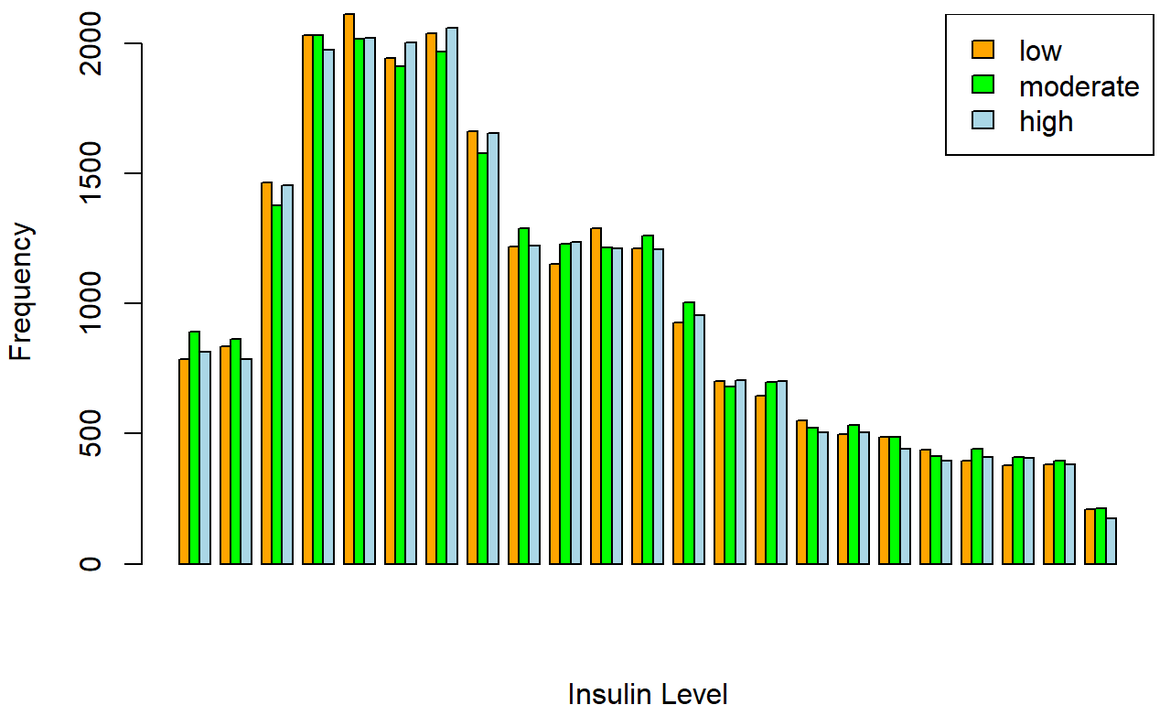

```
summary(diabetes$`Insulin Levels`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5.00   13.00   19.00   21.61   28.00   49.00
```

Na temelju nacrtanog histograma možemo uočiti da više ljudi ima niže razine inzulina (od 0 do 20) te da manji broj ljudi ima razinu inzulina veću od 35.

Usporedba grupiranih podataka:

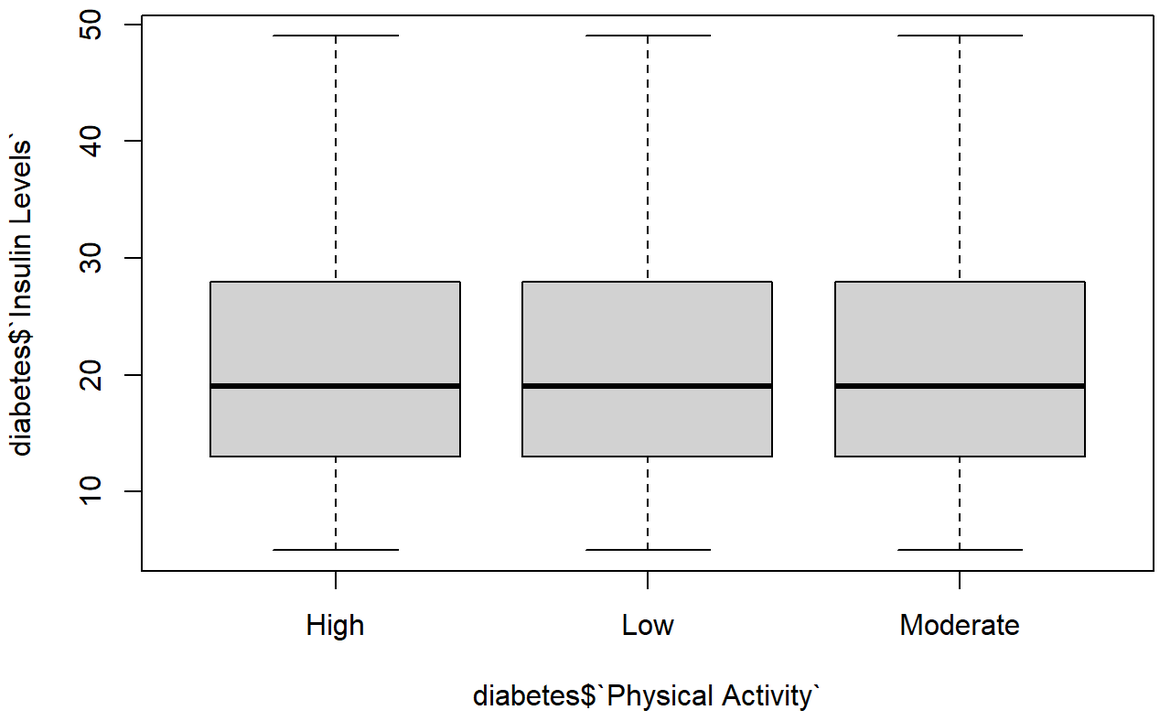
```
# Ako grupiramo podatke i onda radimo histogram:

h1 = hist(diabetes[diabetes$`Physical Activity` == c("Low"),]$`Insulin Levels`,
         plot=FALSE)
h2 = hist(diabetes[diabetes$`Physical Activity` == c("Moderate"),]$`Insulin Levels`,
         plot=FALSE)
h3 = hist(diabetes[diabetes$`Physical Activity` == c("High"),]$`Insulin Levels`,
         plot=FALSE)

data <- t(cbind(h1$counts,h2$counts,h3$counts))
#data
barplot(data,beside=TRUE, col=c("orange", "green", "lightblue"), xlab="Insulin Level", y
lab='Frequency',)
legend("topright",c("low","moderate","high"),fill = c("orange", "green", "lightblue"))
```



Kao

što je vidljivo iz histograma, frekvencije razine aktivnosti ispitanika u ovisnosti o razini inzulina nemaju velikih odstupanja unutar grupa.

```
#boxplot(`Insulin Levels` ~ `Physical Activity`, data=diabetes)
boxplot(diabetes$`Insulin Levels` ~ diabetes$`Physical Activity`)
```



Boxplot koji nam na još jedan grafički način prikazuje raspodjelu ispitanika u ovisnoti o razini inzulina i fizičke aktivnosti.

Grafički prikazi do sad nam daju naslutiti da nema zavisnosti između fizičke aktivnosti i razine inzulina u krvi ispitanika.

Provođenje ANOVE

Pretpostavke ANOVE su:

- nezavisnost pojedinih podataka u uzorcima
- normalna razdioba podataka
- homogenost varijanci među populacijama

```
require(nortest)
```

Lillieforsov test provodimo kako bismo proverili jesu li naši podatci koje proučavamo normalno distrbuirani

```
#provjera distribucije cijelog atributa `Insulin Levels`
lillie.test(diabetes$`Insulin Levels`)
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  diabetes$`Insulin Levels`
## D = 0.11445, p-value < 2.2e-16
```

```
#provjera distribucije atributa `Insulin Levels` = "Low"
lillie.test(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Low"])
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Low"]
## D = 0.11711, p-value < 2.2e-16
```

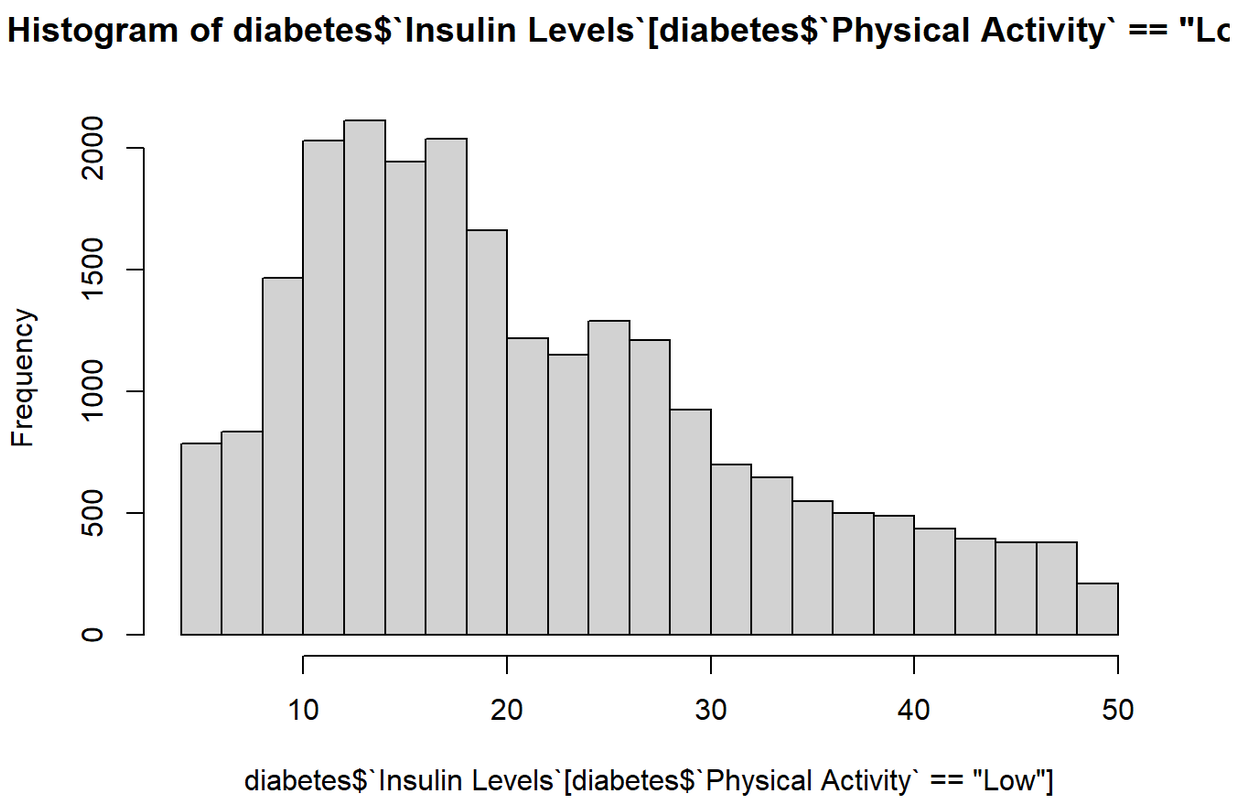
```
#provjera distribucije atributa `Insulin Levels` = "Moderate"
lillie.test(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Moderate"])
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Moderate"]
## D = 0.11076, p-value < 2.2e-16
```

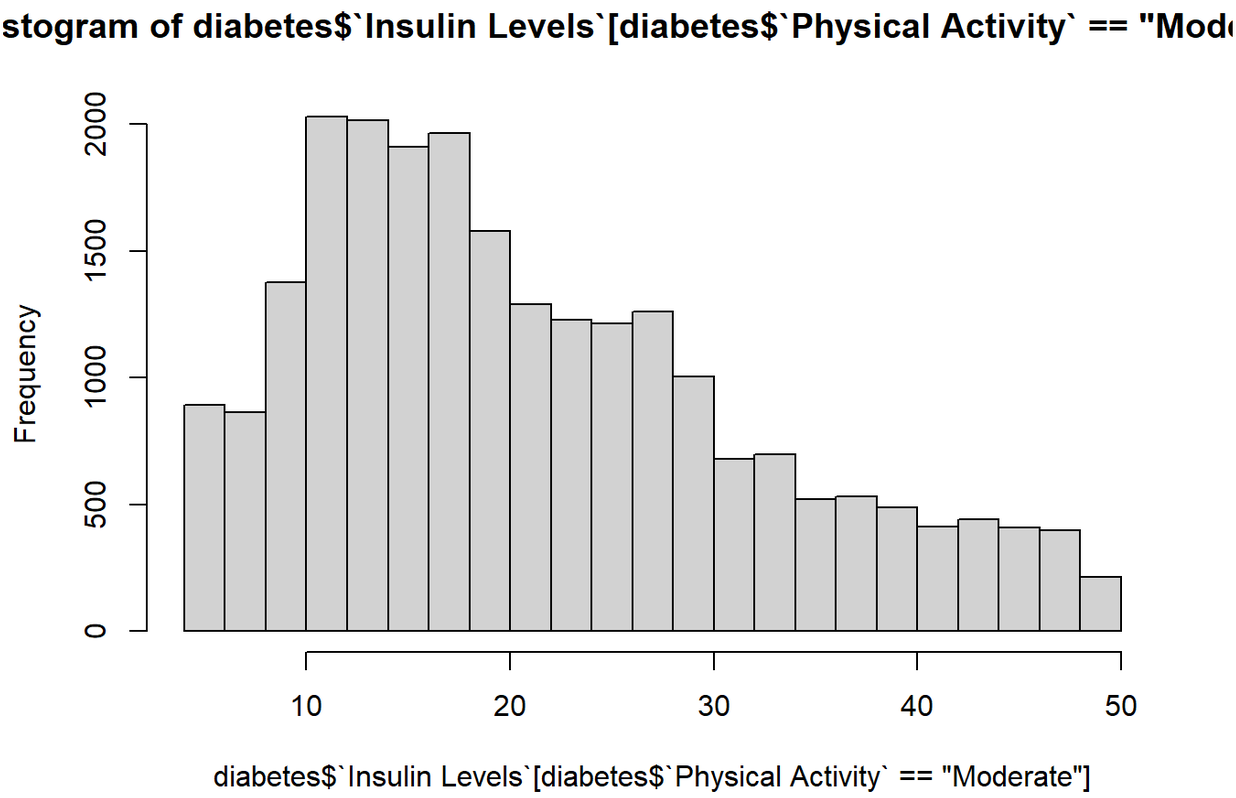
```
#provjera distribucije atributa `Insulin Levels` = "High"
lillie.test(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "High"])
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "High"]
## D = 0.11548, p-value < 2.2e-16
```

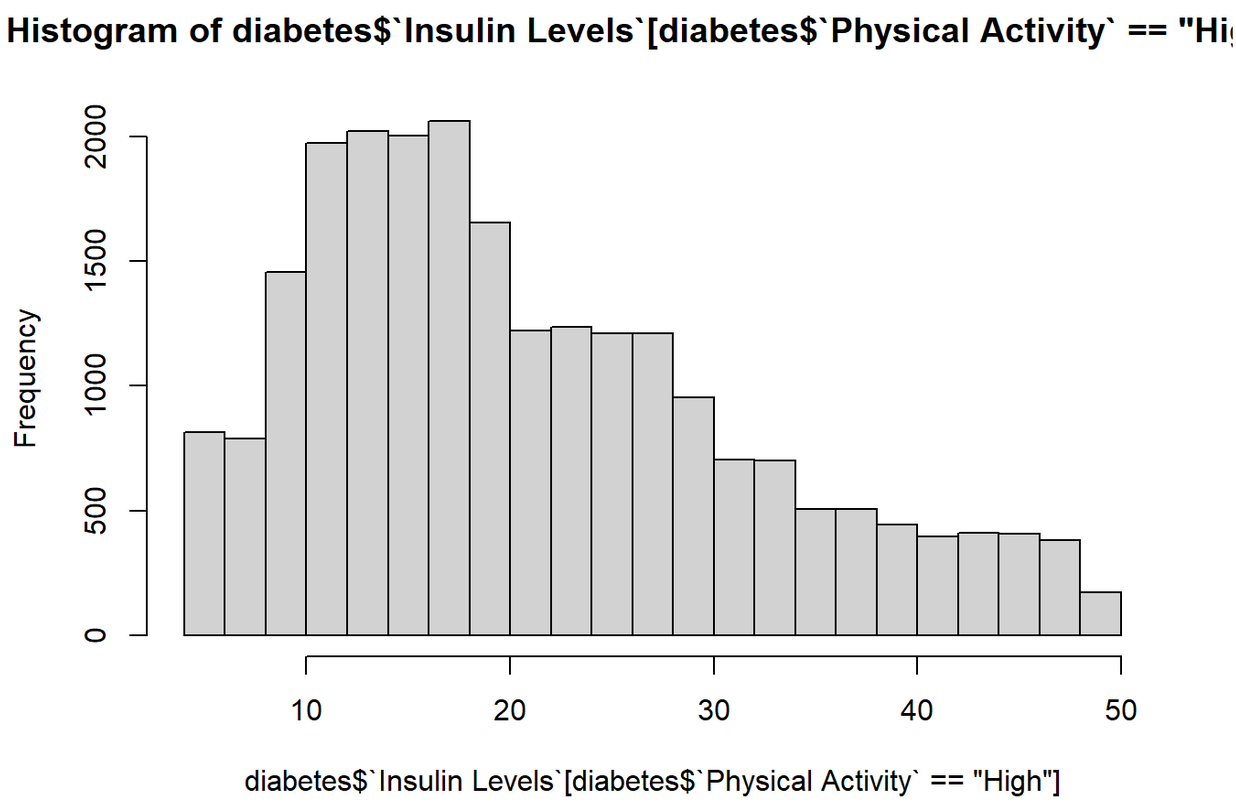
```
#Grafički prikaz podataka
hist(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Low"])
```




```
hist(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Moderate"])
```




```
hist(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "High"])
```




```
# Testiranje homogenosti varijance uzoraka Bartlettovim testom
bartlett.test(diabetes$`Insulin Levels` ~ diabetes$`Physical Activity`)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  diabetes$`Insulin Levels` by diabetes$`Physical Activity`
## Bartlett's K-squared = 6.4479, df = 2, p-value = 0.0398
```

```
var(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Low"])
```

```
## [1] 115.882
```

```
var(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "Moderate"])
```

```
## [1] 118.4482
```

```
var(diabetes$`Insulin Levels`[diabetes$`Physical Activity` == "High"])
```

```
## [1] 114.6527
```

```
a = aov(diabetes$`Insulin Levels` ~ diabetes$`Physical Activity`)
summary(a)
```

```
##                                 Df  Sum Sq Mean Sq F value Pr(>F)
## diabetes$`Physical Activity`     2     346   173.1   1.488  0.226
## Residuals                    69997 8142960   116.3
```

Kako podatci *ne ispunjuju zahtjeve nužne za provođenje ANOVE; ne ispunjuju zahtjev normalnosti, niti homogenosti* (jer su pri provođenju testova p-vrijednosti bile manje od 0.05).

Zato *provodim Kruskal-Wallisov test* jer on ne zahtjeva normalnu distribuciju podataka.

```
# Kruskal-Wallis
kruskal.test(diabetes$`Insulin Levels` ~ diabetes$`Physical Activity`)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  diabetes$`Insulin Levels` by diabetes$`Physical Activity`
## Kruskal-Wallis chi-squared = 2.0217, df = 2, p-value = 0.3639
```

Zaključak:

P-vrijednost dobivena Kruskal-Wallis testom iznosi 0.3639 te veća od 0.05, što znači da ne možemo odbaciti nultu hipotezu(H0): Nema statistički značajne razlike među grupama fizičke aktivnosti u razini inzulina u krvi. Što se dalo naslutiti i grafičkim prikazima, ali nam je numerički ovako potvrđeno.

Istraživačko pitanje

Postoji li razlika u godinama otkrivanja dijabetesa tipa 1 i tipa 2?

```
sum(is.na(diabetes$Target)) # nema NA vrijednosti
```

```
## [1] 0
```

```
sum(is.na(diabetes$Autoantibodies)) # nema NA vrijednosti
```

```
## [1] 0
```

```
#diabetes

diabetes$Age <- as.integer(diabetes$Age) # je li treba mijenjati podatke u integer ??
table(diabetes$Target)
```

```
##
##    Cystic Fibrosis-Related Diabetes (CFRD)
##                                       5464
##                      Gestational Diabetes
##                                       5344
##                                       LADA
##                                       5223
##                                       MODY
##                                       5553
##           Neonatal Diabetes Mellitus (NDM)
##                                       5408
##                                 Prediabetic
##                                       5376
##                        Secondary Diabetes
##                                       5479
##                   Steroid-Induced Diabetes
##                                       5275
##                           Type 1 Diabetes
##                                       5446
##                           Type 2 Diabetes
##                                       5397
## Type 3c Diabetes (Pancreatogenic Diabetes)
##                                       5320
##                  Wolcott-Rallison Syndrome
##                                       5400
##                           Wolfram Syndrome
##                                       5315
```

```
# diabetes


# filtriranje redova koji sadrze samo zapise o tipu 1 ili 2 dijabetesa
type_1_and_2 <- diabetes %>% filter(Target %in% c("Type 1 Diabetes", "Type 2 Diabetes"))

# provjera - proucavanje filtiranih redova - izvlacimo nasumicnih 10
random_rows <- type_1_and_2[sample(1:nrow(type_1_and_2), size = 10, replace = FALSE), ]
print(random_rows)
```

```
## # A tibble: 10 × 34
##    Target          `Genetic Markers` Autoantibodies `Family History`
##    <fct>           <fct>             <fct>          <fct>
##  1 Type 1 Diabetes Negative          Negative       No
##  2 Type 1 Diabetes Positive          Negative       Yes
##  3 Type 1 Diabetes Positive          Negative       Yes
##  4 Type 1 Diabetes Negative          Negative       No
##  5 Type 2 Diabetes Negative          Negative       Yes
##  6 Type 1 Diabetes Negative          Negative       No
##  7 Type 2 Diabetes Positive          Negative       Yes
##  8 Type 2 Diabetes Negative          Negative       Yes
##  9 Type 1 Diabetes Positive          Negative       No
## 10 Type 1 Diabetes Negative          Negative       No
## # ℹ 30 more variables: `Environmental Factors` <fct>, `Insulin Levels` <dbl>,
## #   Age <int>, BMI <dbl>, `Physical Activity` <fct>, `Dietary Habits` <fct>,
## #   `Blood Pressure` <dbl>, `Cholesterol Levels` <dbl>,
## #   `Waist Circumference` <dbl>, `Blood Glucose Levels` <dbl>, Ethnicity <fct>,
## #   `Socioeconomic Factors` <fct>, `Smoking Status` <fct>,
## #   `Alcohol Consumption` <fct>, `Glucose Tolerance Test` <fct>,
## #   `History of PCOS` <fct>, `Previous Gestational Diabetes` <fct>, …
```

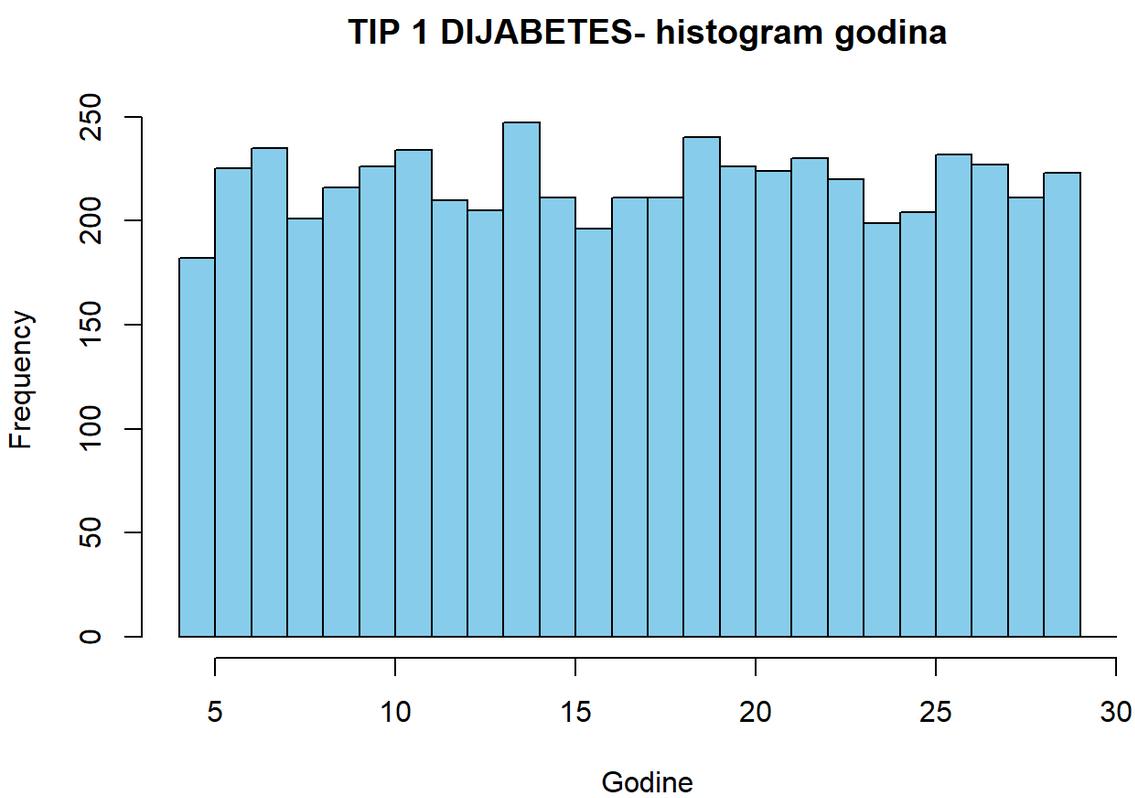
```
# histogram godina ljudi oboljeli od tip 1 dijabetesa
type_1 = type_1_and_2[type_1_and_2$Target=="Type 1 Diabetes",]
type_2 = type_1_and_2[type_1_and_2$Target=="Type 2 Diabetes",]
```

Uočavamo na histogramima da godine(dob) nisu normalno raspoređene.

```
# kako postavit granice historaga
cat('Prosječna dob otkrivanje tip 1 dijabetesa je ', mean(type_1$Age),'\n')
```

```
## Prosječna dob otkrivanje tip 1 dijabetesa je  17.065
```

```
hist(type_1$Age,
     breaks = seq(min(type_1$Age) - 1, max(type_1$Age) + 1, 1),
     main = 'TIP 1 DIJABETES- histogram godina',
     xlab = 'Godine',
     col = 'skyblue',
     border = 'black')
```



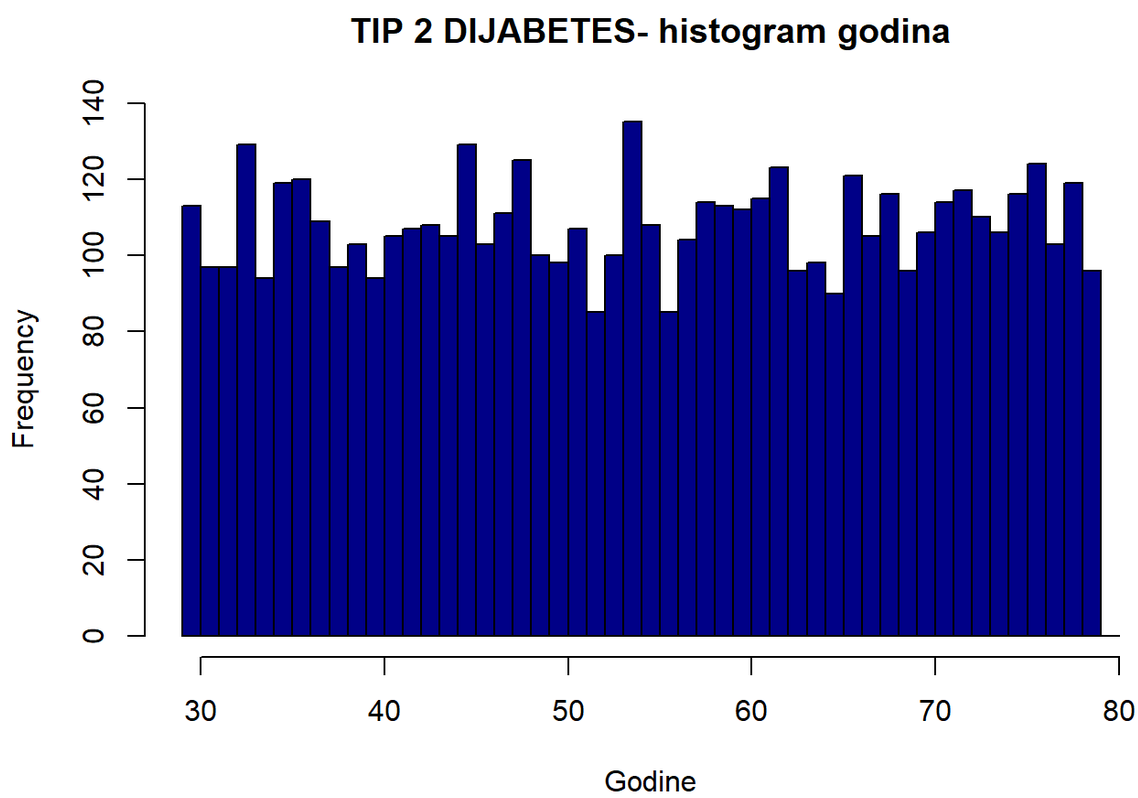

```
summary(type_1$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5.00   11.00   17.00   17.07   23.00   29.00
```

```
cat('Prosječna dob otkrivanje tip 2 dijabetesa je ', mean(type_2$Age), '\n')
```

```
## Prosječna dob otkrivanje tip 2 dijabetesa je  54.60978
```

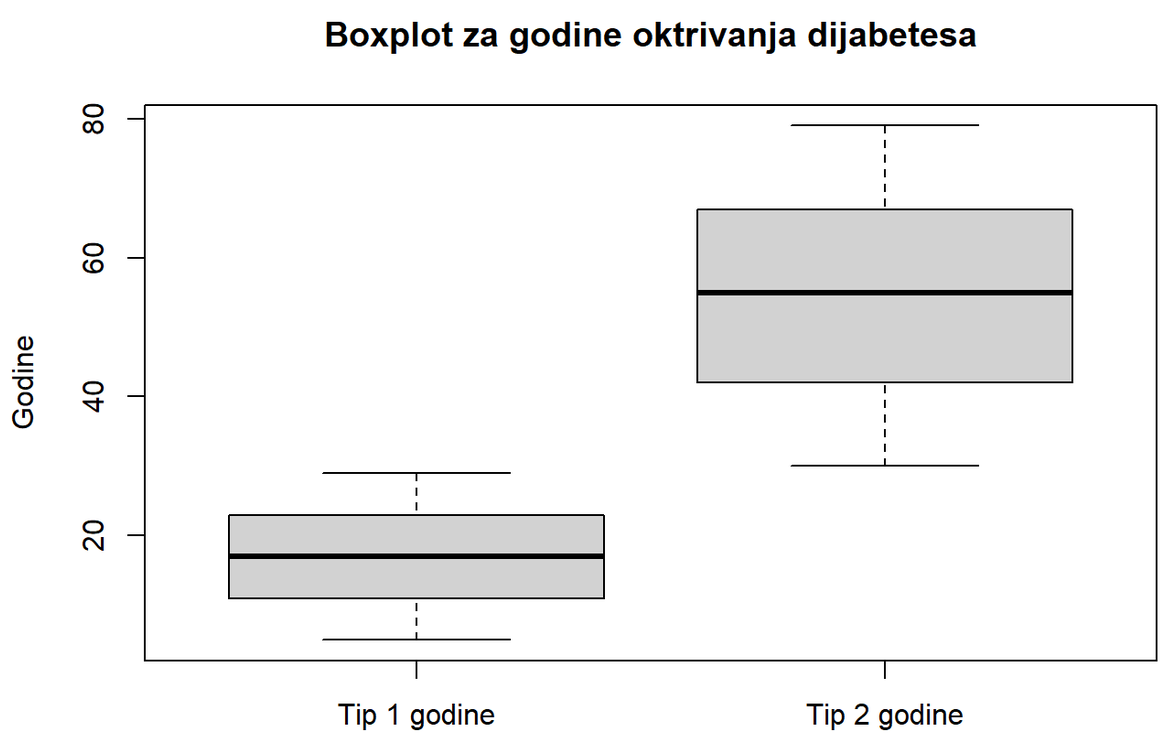
```
hist(type_2$Age,
     breaks = seq(min(type_2$Age) - 1, max(type_2$Age) + 1, 1),
     main = 'TIP 2 DIJABETES- histogram godina',
     xlab = 'Godine',
     col = 'darkblue',
     border = 'black')
```



Uočavamo da podaci godina(dob) nisu normalno distribuirani nego jednoliko.

Pomoću boxplot dijagrama ćemo nacrati da bi se razlika jasnije vidjela.

```
boxplot(type_1$Age, type_2$Age,
        names = c('Tip 1 godine','Tip 2 godine'),
        main='Boxplot za godine oktrivanja dijabetesa',
        ylab="Godine"
        )
```



Zaključak:

Iz grafičkih rezultata jasno uočavamo da je otkrivanje dijabetesa tipa 1 puno ranije nego dijabetesa tipa 2